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A simple approach for adding thixotropy to an elasto-visco-plastic rheological model to facilitate structural interrogation of human blood

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ABSTRACT

We demonstrate refinements of a simple elastic-viscous-plastic rheological model with the addition of a thixotropic term to better model rheological flow of human blood. The alteration of this model will contribute to better representation of *lightly* structured complex materials. The simple elastic-viscous-plastic (EVP) framework requires thixotropy for more accuracy in fitting and predicting the thixo-elasto-visco-plasticity of rheological flows with a microstructure. We demonstrate the performance of the added thixotropic features by comparing the TEVP construct to similar peer models using a cost function generated from predictions of rheological flows from triangle ramps to large amplitude oscillatory shear flow. To verify the efficacy of these thixotropic enhancements, the performance of the model is then measured against the other models with a penalty assessed based on number of model parameters. The framework holds the name "Enhanced Thixotropic-Elasto-Visco-Plastic" to reflect this methodology of adding thixotropy to the elastic-viscous-plastic capabilities of the standard construct.

1. Introduction

Blood rheology, or the study of viscous, elastic, plastic, and thixotropic properties of blood, is rapidly becoming a topic of interest in diagnosing various vascular-related disorders [1–4] and modeling the rheology of human blood has shown an increase in relevance and popularity [5–9]. Researchers have historically targeted modeling the rheological properties of blood through continuous refinement of previous models. Simple models tend to target steady state flow of blood while the more advanced and dynamic models target the more complex transient flow induced microstructural transitions of blood and the viscoelastic properties of blood. The advanced models can more accurately model the thixotropic, elasto-visco-plastic behavior of blood because blood acts as a non-Newtonian fluid, meaning the viscosity changes based on not only the shear rate but also the level of microstructure due to rouleaux [1–9]. By incorporating a thixotropic addition properly, we will improve the ability of the model to represent the transient flow of blood, and simultaneously predict the microstructure level.

Blood has a multi-faceted role in the human body, serving as a control for a multitude of functions in the body: performing mass transfer processes of varying substances (i.e. oxygen, carbon dioxide, nutrients, etc.), regulating body heat, and maintaining homeostasis [1–6]. The shear rate (velocity) of blood determines the flow induced behavior of microstructure, including the rate of formation, and breakdown of microstructure or rouleaux (red blood cells can form *'coin-like'* stacks). The rouleaux in turn contribute to the overall viscosity of the blood. The complexity arises because blood is classified as a liquid suspension, plasma, with solids within: red blood cells, white blood cells, platelets, and cholesterol [1–4]. The proteins within the plasma of the blood play an important function by maintaining the osmotic pressure of plasma. The blood has proteins and contains fibrinogen which helps blood clot, preventing major blood loss due to damage to the blood vessel walls [1–9]. Erythrocytes (red blood cells) carry oxygen from the lungs throughout the body, subsequently transporting carbon dioxide for a resupply of oxygen from the lungs. Leukocytes (white blood cells) contain a nucleus, granules, and mitochondria, all surrounded by the cytoplasm. Platelets, like fibrinogen, are important for blood clotting in damaged blood vessels [2]. Early experiments have revealed that the viscosity of blood is strongly dependent upon the applied strain and shear rate and the physiological parameters of the hematocrit, fibrinogen, and cholesterol levels [10]. The viscosity and elasticity of blood is also dependent upon the size of the tubes (veins; arteries) in which it flows, altered by ventricular contraction and expansion [3,11].

Traditional models of fluid and rheological flow focus on the elastic,

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Nomenclature

ω	frequency of oscillation	k_λ	thixotropic time constant of evolution
γ_0	strain amplitude	σ	stress
t	time	γ	shear rate
T	temperature	a	power law of shear breakdown
λ	structure parameter	d	power law of shear aggregation
η_{st}	structural contribution to viscosity	e	elastic component
η_∞	infinite shear viscosity	p	plastic component
G'	storage modulus	y_i	data value
G"	loss modulus	f_i	model value
G	elastic modulus	F_{cost}	cost function
σ_{y0}	yield stress	K	number of model parameters
k	consistency parameter	α	slope of increase in shear rate of triangle ramp
n	power law	D_i	fitting parameters
tr_1	thixotropic time constant of shear breakdown	C_i	fitting parameters for multilambda model
tr_2	thixotropic time constant of shear buildup	B	fitting parameter
		A	back strain structure parameter
		τ_{ii}	stress tensor components in *ii* direction

viscous, and plastic characteristics of the flow. The Herschel-Bulkley model, for example, only targets two features using a linear superposition of a varying viscous stress, and a yield stress [12]. In general, an increase in shear rate results in a decrease in viscosity for human blood. The most common shortfall with the traditional models is that they lack the ability to effectively model the dynamic and simultaneously changing rheological properties like viscosity and microstructure. The Herschel-Bulkley model cannot effectively model the changes in viscosity and microstructure induced by changing shear rate, to include build-up and break-down of structure during transient rheological flow [12,13]. Stickel et al. [14] identified a simplified model for elasto-visco-plastic materials using a "Maxwell-like representation of the steady shear Herschel-Bulkley model", based on a previous and similar model by Saramito [14,15]. This model accounts for non-zero changes in shear stress, as well as a finite elastic modulus, *G*. Large amplitude oscillatory shear (LAOS), whereby strain is given as follows, $\gamma = \gamma_0 sin(\omega t)$, and the first time derivative, shear rate is $\dot{\gamma} = \omega\gamma_0 cos(\omega t)$ where ω is the frequency of oscillation in (radians/s) and γ_0 is the strain amplitude (dimensionless), were used by Stickel *et al.* to demonstrate the efficacy and flexibility in one of the original forms of the model [14].

Separately, Dullaert and Mewis, Mujumdar *et al.*, Mewis and Wagner, de Souza Mendes and Thompson, Blackwell and Ewoldt, Apostolidis *et al.*, Armstrong *et al.*, and Horner *et al.* incorporated and improved upon a thixotropic framework that utilized three general modes of structure dynamics: shear structure breakage; and structure rebuilding due to Brownian forces and shear aggregation [7–9, 12–22]. In addition, novel work on modeling thixotropy was presented by Wei and Solomon that incorporates this generalized framework, among microstructure modeling techniques [23,24], including a term for 'back-stress', that comes from the microstructure of the material. The framework incorporated here can model and predict microstructure in both steady state and transient conditions by accurately representing structure build up and breakdown from changes in shear rate. The model includes λ to represent a structure parameter in this model. A value of *1* for λ represents a fully structured material, whereby all connections and bonds with nearest neighbors are intact, while a value of $\lambda \rightarrow 0$ represents complete breakdown, with close to zero microstructure [12,13,19]. In a sense, this model more accurately accounts for microstructure in complex materials that are subject to deformation upon changes in shear rate. This deformation is the basis of viscoelastic changes because of changing shear rate, thusly impacting the stress measurement [5,8,9,13, 19,20]. For this model, a change in structure results in a change in elasticity as well, which requires a correction to the yield stress and elastic modulus as shown here [5,14,15]. Therefore, for this thixotropic model, total stress is a combination of contributions from the solvent background and the microstructure. In addition, Dullaert and Mewis, as well as Mewis and Wagner distinguished between two types of viscosity in a dynamic state: viscosity due to variable microstructure, $\lambda\eta_{ST}$ and pure solvent viscosity, η_∞ [12,13]. By distinguishing different contributions to the viscosity, Mewis and Wagner, and Dullaert and Mewis effectively modeled the variation in viscosity in a transient, or changing shear rate condition [12,13].

Several modifications to this framework were explored by Dimitriou et al. [25,26] in development of the IKH model. The IKH model added back stress to account for kinematic hardening. To do this the strain and shear rate were separated into elastic and plastic components, and the total stress became the product of the elastic modulus and elastic strain. In addition, Dimitriou et al. added a dimensionless parameter, *A* to account for an internal variable which corresponds to specific microstructure of the material. When the material is not at steady state Dimitriou et al. added an ordinary differential equation to describe the evolution of parameter *A*. The model included a 3D tensor version. Following this effort, Dimitrou and McKinley introduced a model that included thixotropic features as well as dimensionless variable *A* for crude oil [26].

Further modifications were made to the model with a framework by Wei et al. and Varchanis et al. using a silica suspension that simultaneously utilized thixotropy as well as dimensionless internal variable *A* [23,27]. Wei et al. also broke the total stress into a contribution from the medium, and the structure. Using this paradigm Wei et al. made several modifications to the constitutive equation using elements of the evolving structure including yield stress, and structural viscosity. In addition, Wei et al. also added a thixotropic term that included a superposition of different "sub-structure parameters" with different time constants of evolution, to give the effect of a spectrum of microstructural time evolutions [23,24] (the Wei et al. model, ML-IKH is shown Supplemental Material). The ML-IKH model uses back stress to calculate the effective stress, which is the quantity incorporated into the differential equation for structural contribution to total stress. This is where the work of Varchanis et al. begins. Varchanis et al. completely left out the medium contribution to total stress and built a fully tensorial model based only on the contributions from the evolving microstructure, with a full tensor to describe ***A*** the dimensionless variable describing features relevant to the specific microstructure of the material, called the SPPT-IKH (model framework shown in Supplemental Material). SPPT-IKH used the product of current structure and structural viscosity in the differential equation for the structural contribution to total stress. We point out here that human blood has a significant portion of its total stress contribution from the medium (plasma) that the SPTT-IKH model is not appropriate for modeling blood [27], however we incorporate the

$\lambda\eta_t$ term in version *2* of our proposed model construct, without the back stress term for best results modeling human blood.

Additionally, there has been much progress in modeling thixo-elasto-visco-plasticity contributed by de Souza Mendes and coworkers recently. De Souza Mendes first introduced a TEVP model based on the Maxwell framework, with an evolving viscosity and elastic modulus function in which both were functions of microstructure via the structure parameter, λ [16]. Following this effort, de Souza evolved his modeling by incorporating an Oldroyd-B, or Jeffreys approach for the backbone of the model, and still utilized the evolving microstructure term with overall viscosity and elastic modulus as a function of current value of *lambda* [17]. Following these contributions, de Souza Mendes and Thompson modified the thixotropic model while simultaneously maintaining a structure build-up and breakdown term [18]. The sequence of contributions by de Souza and coworkers was a major step forward in the development of thixo-elasto-visco-plastic modeling, and the advances were made in part by incorporating the baseline Maxwell and Jeffreys frameworks with thixotropic modifications [16–18].

Previous rheological tests have been performed examining various components of blood as the baseline parameters to gather data on how blood performs under varying shear rates. For example, a group of researchers examined the effect of cholesterol and triglycerides on blood rheology under shear [5,6,28–30]. They did this by using samples of blood with low and high levels of cholesterol. Through this research they were able to find links between cholesterol/triglycerides and cardiovascular diseases. Additional experiments have been conducted to prove that a model works in predicting time-dependent shear flow rheology of blood at low and moderate values of shear rate [5,6,28–30]. Models have been interrogated or tested with several common rheological tests such as triangular steps and rectangular step-up and step-down as was done by Bureau et al. [31].

Oscillatory rheological experiments of both small and large amplitude have been extensively employed to interrogate complex materials, and we use results from such experiments to validate model enhancements here [8,9,28,30]. The linear regime is probed by SAOS, where SAOS is the linear regime whereby the two parameters storage modulus, *G'* (Pa) and loss modulus, *G"* (Pa) can fully reconstruct the stress signal as follows: $\sigma(t) = \gamma_0[G' sin(\omega t) + G'' cos(\omega t)]$. In other words, there is only a single harmonic, consisting of two moduli required to reproduce the stress signal in SAOS. Responses to large amplitudes cannot be accurately described by two parameters only, leading to representations that include multiple harmonics [29,30], and more recently the transition to medium amplitude oscillatory shear flow (MAOS) from SAOS has been actively studied. Several groups modified a thixotropic suspension model to include using LAOS predictions as a metric of model efficacy [21,22,28]. Similar rheological experiments using LAOS have been conducted using varying materials with a microstructure, to include various suspensions and human blood [5,8,9,21–24, 30]. The rheological properties of various suspensions are different than those of blood; however, thixotropic model enhancements have increased robustness of model testing, verification, and validation for all suspensions with a microstructure. This in turn facilitates a better understanding of the microstructure of complex material including human blood and is useful for human blood rheologists. Researchers were able to validate and improve constitutive models of thixotropy and test and validate modernized structure-kinetics models [22]. Through the human blood rheological modeling, researchers past and present were able to identify weaknesses and limitations of thixotropy models and further apply this knowledge to rheological blood study [12,13,21–23].

In experiments under steady and unidirectional-LAOS, where $\dot{\gamma} = \omega\gamma_0 cos(\omega t) + \omega\gamma_0$ (a linear superposition of a steady shear rate and an oscillating contribution such that the rheometer will not rotate in the negative direction) the model predicts the stress responses of blood donors using a range of strain amplitudes and frequencies [8]. This type of rheological experiment is performed to better understand the arterial pulsatile blood flow. Several rheologists used modern thixotropic-viscoelastic models and compared their modifications to legacy models using steady state, transient, and oscillatory shear flow experiments [9,21,22]. Additional research has been conducted to create user friendly models, specifically for non-linear models where data-driven parameter estimates consume the research.

Table 1
Donor physiological test results.

Donor	Hematocrit (%)	Fibrinogen (mg/dL)	Total Chol. (mg/mL)	Triglycerides (mg/mL)	HDL Chol. (mg/mL)	LDL Chol. (mg/mL)
1	42.9	0.242	138	108	35	83
2	46.2	0.316	132	111	41	72
3	38.3	0.333	145	144	39	82
4	45.2	0.214	187	57	65	108

This manuscript defines, demonstrates, and then compares a modified framework that combines Stickel and coworkers' elasto-visco-plastic only model with legacy thixotropic modeling to describe the transient effects of complex materials with microstructure more accurately [12–14, 19,21,22]. For blood rheology, the microstructure is contained in the plasma that contains red blood cells. Instead of only modeling steady-state blood with one term to represent the effects of changing viscosity, as the original Herschel-Bulkley model did, the thixotropy-modified, proposed model will more effectively capture the stress, everchanging structure, and rheology of transient flow of blood. By doing so, the model will show the effects of changes to shear rate and structure [12,13,19,21,22]. The transient flow regime, where structure is actively evolving, will be captured through the addition of a thixotropic term, a non-zero, elastic modulus, *G*, and an evolving yield stress σ_{y0}. The σ_{y0} term will be modified with current structure level *lambda*. With this addition, the model will more effectively capture the decreased structure and viscosity because of increased shear rate [13,24, 26,27]. This proposed model also includes η_{ST}, with λ multiplier, to model the variation of impacts from microstructure and structure breakdown, respectively on the structural viscosity contribution, as done in tensorial form in the SPTT-IKH model. With a thixotropic addition to the scalar elasto-visco-plastic model, one can more effectively model nonlinear and transient rheological flows for human blood models using CFD software, as well as determine non-ideal physiological traits of blood because of deviation from ideal blood parameter values (i.e. increased infinite shear viscosity, increased value of yield stress) [5,6, 10,29–31].

The addition of thixotropy to the EVP only model from Stickel et al. [14] will enable a more nuanced study of the transient rheological data of human blood and more robust model to perform parametric and statistical analysis with. In part, our goal is to use the thixotropic enhancements to correlate the elastic and viscous properties evolution directly to our model predictions of structure. This will validate our model's additions and tie the structure to the evolving elastic and viscous nature of human blood, as there is an elastic and viscous contribution to stress from the microstructure. The remainder of the manuscript is as follows: materials and experimental protocol; description of model enhancements; model parameter fitting procedure; results; and conclusions.

2. Materials and experimental protocol

Four healthy donors were sampled. While sitting, blood was drawn from the antecubital vein through a 21 G needle after the application of a tourniquet, into a 6 mL Vacutainer tube containing 1.8 mg/mL of ethylenediaminetetraacetic acid. Each of the donors had previously fasted for 8 – 10 hours [8,9,30]. The protocol is consistent with guidelines established by the International Society for Clinical Hemorheology [8,9, 33]. Additional samples from the same donors were collected and subsequently sent to Quest Diagnostics for Complete Blood Count, Lipid

Panel, and Fibrinogen Activity testing [8,9,33,34].

The first sample from each donor was brought directly to the ARES-G2 rheometer, while the remaining samples were sent for testing [8,9]. The results of the physiological tests for each of the donors are summarized in Table 1. All the rheological data was taken with the ARES-G2 rheometer equipped with a double wall Couette geometry [8,9]. This rheometer can measure shear rates over a wide range and has a minimum torque and angular frequency of 0.1 N m and 1 rad/s, respectively [8,9]. The geometry was selected to maximize contact surfaces and the vertical orientation while minimizing interfacial area and required loading volume [8,9]. The bob is composed of titanium and the cup of stainless steel, materials that have been shown to have insignificant interactions with blood [8,9,31]. For the inner wall, the measurement gap is 0.43mm, and the measurement gap for the outer wall is 0.5mm. The required loading volume for this geometry is 5ml. All blood samples were measured at 37°C. This temperature was maintained using a Peltier temperature controller with a sensitivity of 0.1°C. To minimize effects of aging, all measurements were completed within 4 hours of withdrawal [8,9]. Prior to measuring the samples, the blood underwent a transportation period of 45–60 minutes during which the sample was stored at room temperature. The sample was loaded into the measurement geometry using a 5ml syringe equipped with a 21 G needle [8,9,31]. A solvent trap was used to minimize effects of evaporation, and the sample was allowed 5min to equilibrate to the temperature before testing. To prevent damage to the red blood cells, shear rates within the rheometer never exceeded 1000 s^{-1}. A preshear of 300 s^{-1} for 30s was implemented between subsequent tests to remove any memory effects from the previous test [8,9,31]. We note here that we did not test different geometries with respect to human blood for this work to test for issues from surface tension torque and interfacial protein film, and hereby acknowledge this as a limitation. Note that although several donors were sampled, only results for donor 3 are graphically shown in the main text, however model fit results are included for the three addition donors in the Supplemental Material [8,9,31]. We acknowledge that this is a limitation of the work.

3. Proposed models and enhancement descriptions

The basis for the model stems from identifying the components of human blood (complex material that acts like a suspension) that make up the total stress of the system, which is the force required to produce the velocity gradient or the shear rate.

$$\sigma_{tot} = \sigma_{structure} + \sigma_{medium} \tag{1}$$

In Eq. (1), σ_{tot} represents the total stress of the structure contribution, and the medium. Essentially, the total stress is composed of the sum of the stress from the medium, which is plasma in the case of blood, and the various contributions to stress from the structure of the blood (red and white blood cells, proteins, etc.) [12–15].

The basic model structure that set the stage for advances in rheological models and is a standard for viscoplastic rheological modeling is the work of Saramito [15]. This evolved into work from Stickel et al. [13,14], Dimitriou et al. [25,26], Wei et al. [23] and Varchanis et al. [27]. In each case the Herschel-Bulkley Model is the starting point [13,14],

$$\sigma_{tot} = \sigma_{y0} + k\dot{\gamma}^n \tag{2}$$

where $\dot{\gamma}$ represents shear rate, σ_{tot} represents shear stress, σ_{y0} represents yield stress, k is the consistency parameter, and n is the power law parameter that describes whether a fluid is shear thinning or shear thickening. Using hematocrit and fibrinogen data from Table 1, the yield stress, and infinite shear viscosity can be calculated as follows from [5,6]

$$\sigma_{y0} = \left(\left(HCT - \left(0.3126 \times c_f^2 - 0.468 \times c_f + 0.1764\right)\right)^2\right), \quad \left(0.5084\left(c_f + 0.4517\right)^2\right) \tag{3}$$

$$\eta_\infty = n_p\left(1 + 2.0703 \times HCT + 3.7222 \times HCT^2\right)\exp\left\{-7.0726\left(1 - \frac{T_0}{T}\right)\right\}. \tag{4}$$

Where HCT is hematocrit level, c_f is fibrinogen, n_p is the plasma viscosity (taken to be 1.67E-3 Pa s) and T_0 and T represent the refence temperature and temperature at which the data is taken. Eqs. (3) and (4) are used consistently throughout to calculate the yield stress, and when appropriate the infinite shear viscosity [6,29].

In Eq. (2) the elastic stress is accounted for with the single term for yield stress, σ_{y0}, and the viscous component of the stress accounted for with $k\dot{\gamma}^n$. When $n > 1$, the fluid is shear thickening at high shear rates, while shear thinning at low shear rates. When $n < 1$, the fluid is shear thinning [13,19,23,30]. Herschel-Bulkley is specifically good for steady state modeling. It does not consider the microstructure in fluids and the plasma of blood separately. Eq. (2) allows for the computation of the stress based on the shear rate and is used to fit the steady state data.

The goal of this effort is to construct a model that progresses from modeling "steady state" to model the transient rheology of human blood by accounting for blood's microstructure [8,9] with the least number of model parameters. The Herschel-Bulkley model can capture the steady-state flow curve rheology of many materials but has been shown to struggle with accurate fitting and modeling of transient rheological experiments. An elasto-visco-plastic (EVP) like representation of steady state provided by the Herschel-Bulkley is a starting point to capture transient rheology data by considering the elastic modulus. This allows one to set a characteristic 'viscoelastic' time scale of evolution of certain fluids, such as the plasma of blood, as seen in Eq. (5). In Eq. (5), the only new parameter is G, which is the elastic modulus and is a measure of stiffness of a fluid resulting from microstructure, or polymer entanglements, etc. [12–15]

$$\frac{1}{G}\dot{\sigma} + \max\left(0, \frac{|\sigma| - \sigma_{y0}}{k|\sigma|^n}\right)^{1/n}\sigma = \dot{\gamma}. \tag{5}$$

Our proposed model enhancement will incorporate a simple scalar thixotropic term to use as a basis for a transient model. This will include shear breakdown as well as a Brownian and a shear aggregation, or 'build-up' of microstructure as was done by Wei et al, and Varchanis et al. From Stickel et al. we know that Eq. (5) can model steady state, as well as LAOS (and other transient rheological flows) by accurately modeling the viscoelastic time scale through mathematical incorporation of $1/G$ into the dynamic model for some materials. Under steady state conditions, Eq. (5) reverts back to Eq. (2). Eq. (2) is the original Herschel-Bulkley model. Eq. (5) does not have a thixotropic term, or characteristic times of structure evolution. However, human blood requires not only a viscoelastic time scale, but also the thixotropic timescales for the contributions of the microstructure formed by the rouleaux, to adequately be accounted for [8,9,14,15]. Thus, to Eq. (5) we will add the structure parameter *lambda* as a multiplier to the yield stress, and the thixotropic viscous element, η_{ST} (shown in Eq. (11)). This will more accurately represent the evolution of the microstructure seen in human blood.

To capture rheological effects of the microstructure of the material, Dimitriou and coworkers used a different approach. With the Herschel-Bulkley model as a basis, elasticity was incorporated, in the context of one-dimensional deformations, by first decomposing the strain into plastic and elastic components as follows $\gamma_{tot} = \gamma_e + \gamma_p$; and then relating the total stress in the material to the elastic modulus, G as follows $\sigma = G\gamma_e$ [25,26]. The rate of plastic strain is then determined as

$$\dot{\gamma}_p = \begin{cases} 0 \text{if} |\sigma| < \sigma_{y0} \\ n^p\left(\frac{|\sigma| - \sigma_{y0}}{k}\right)^{1/n} \text{if} |\sigma| \geq \sigma_{y0} \end{cases}, \tag{6}$$

with the material constant the same as Herschel-Bulkley, and n_p equal to

the direction of stress, given as $n^p = \sigma/|\sigma|$. This was called the EHB, Elastic Herschel Bulkley. From here, Dimitriou et al. added features to account for kinematic hardening which could account for the movement of the center of location of the yield stress surface for a given material [25,26]. This framework is used to model material plasticity. To fully model the plasticity first Dimitriou et al. defined free energy, $\Psi = 1/2(G\gamma^e)^2 + 1/2C(A)^2$, where A is a dimensionless variable for the 'defect energy'. Dimitriou et al. then defined a back stress, $\sigma_{back} = d\Psi/dA = CA$. Where the back stress in this context refers to the center of the location of the yield stress surface of the material. The direction of plastic flow rate is updated with the back stress as follows

$$\dot{\gamma}_p = n^p\left|\dot{\gamma}_p\right| \text{where} n^p = \frac{\sigma - \sigma_{back}}{|\sigma - \sigma_{back}|}, \tag{7}$$

and the magnitude of the plastic flow rate is now

$$\left|\dot{\gamma}_p\right| = \left(\frac{|\sigma - \sigma_{back}|}{k}\right)^{1/n}. \tag{8}$$

Lastly, the differential equation for the parameter A in the simple 1D case is

$$\dot{A} = \dot{\gamma}_p - qA\left|\dot{\gamma}_p\right|. \tag{9}$$

Furthermore, the parameter q is introduced as a dimensionless, material constant related as follows to yield stress, $\sigma_{y0} = C/q$. This is the IKH model, and more information on this framework is provided here [25]. The simple 1D version of IKH is demonstrated in the results section as a peer-analog, simple EVP model for comparison. The IKH also serves as the basis for the work of Wei et al. and Varchanis et al. [23,25–27]

The shortcoming of the original EVP only model by Stickel *et al.* is that it does not possess the capability to model the dynamic behavior of complex fluids with an actively evolving microstructure like human blood in a transient state over extended periods of time. For a short period of time, the models would be sufficient as they have a viscoelastic time scale, when structure is constant, for example in SAOS flow. This means that the models can represent how the stress-strain relationship in fluids change over time [3,4,14]. However, the models lose their advantage when dealing with long term microstructure evolution scenarios. Such situations are best represented with the presence of the thixotropic time scales. Thixotropy, according to Mewis and Wagner, is "…the continuous decrease of viscosity with time when flow is applied to a sample that has been previously at rest and the subsequent recovery of viscosity in time when the flow is discontinued" [2,12,13]. Without this time scale, the models are incapable of representing a dynamic viscosity when flow changes pace over time.

The IKH was further developed by Varchanis et al. [27] by incorporating a full tensor form for both the stress and dimensionless, parameter A, as well as adding a simple 1D thixotropic parameter, λ, to represent structuring level from [0 1], where the framework will approach *0* (no structure), and *1* is fully structured. The SPTT-IKH model made the stipulation that this version is for materials that have negligible contributions to stress from the solvent, and their original model contains *15* parameters. We show the equations of the SPTT-IKH model, along with demonstration of its capability in the Supplemental Material. Using a similar construct Wei et al. [23] then further enhanced the IKH version of the model by adding a spectrum of thixotropic time constants and continued use of the 'back-stress' term of the original IKH model, calling it the ML-IKH. With the ML-IKH model the shear rate is again broken into plastic and elastic components, with the plastic portion used for the structure evolution equations, and constitutive equation [23,27]. The ML-IKH constitutive equation includes the contribution from the spectrum of structure parameters to the stress differential equation, and back stress together, both a function of the structure parameter. The full ML-IKH model equations are shown in the Supplemental Material.

For a model that more accurately represents the contributions of the microstructure, both elastic and viscous, and solvent contributions simultaneously, as well as their respective time scales of evolution, we incorporate the following from Dullaert and Mewis [13], and Mewis and Wagner [12],

$$\sigma_{ss} = \lambda\sigma_y + \lambda\eta_{ST}\dot{\gamma}_{ss} + \eta_\infty\dot{\gamma}_{ss}, \tag{10}$$

where again, λ represents the current level of microstructure from [0 1], while η_{ST} and η_∞ represent the structural contribution to viscosity and the infinite-shear viscosity, respectively [12,13,19,20]. Eq. (10) is similar to Eq. (2), setting n=*1*; thusly k becomes η_∞, with a viscous contribution from microstructure added, η_{ST}, as well as a λ modifier for both the yield stress and structural viscosity. Eq. (10) is the form of the steady state constitutive equation for both versions of the enhanced EVP model. The steady state flow curve can be fit by Eqs. (10) and (16). For the transient rheological flows when the microstructure is still evolving, we combine Eqs. (5) and (10) as follows (version 1):

$$\frac{1}{G}\dot{\sigma}_{tot} + \max\left(0, \frac{|\sigma_{tot}| - \lambda\sigma_{y0}}{(\lambda\eta_{ST} + \eta_\infty)|\sigma_{tot}|}\right)\sigma_{tot} = \dot{\gamma}, \tag{11}$$

where G is the elastic modulus, and $\lambda\sigma_{y0}$ is the current value of the yield stress based on current microstructure. There is no *lambda* modifier for elastic modulus because it is the contribution from G that locks in the necessary viscoelastic response of the solvent and the structure together time that is necessary for accurate blood fitting and modeling. It is noted here that n as seen in Eq. (5) is set to unity. Additionally, the yield stress term, and the structural contribution to viscosity are multiplied by the current value of the microstructure, *lambda*. The second version of the model, inspired by Varchanis et al. and Wei et al. [23,27] postulates the ordinary different equation represents only the contribution to total stress from the evolving rouleaux

$$\frac{1}{G_R}\dot{\sigma}_s + \max\left(0, \frac{|\sigma_s| - \lambda\sigma_{y0}}{(\lambda\eta_{ST})|\sigma_s|}\right)\sigma_s = \dot{\gamma}_p, \tag{12}$$

where G_R is the elastic modulus of the rouleaux, and the total stress is the linear superposition of the contribution from σ_s, given in Eq. (12) and the contribution from the solvent (in the case of human blood the plasma), $\sigma_m = \eta_\infty\dot{\gamma}$ shown in Eq. (1). Eq. (12) incorporates the plastic shear rate as it directly models the 'plastic process' of stretching the rouleaux, ostensibly in a non-reversible manner. Eqs. (11) and (12) represent the two versions of the constitutive equation, the first inspired by Saramito, Stickel et al., and Dimitriou et al. [14,15,25] and the second inspired by Varchanis et al. and Wei et al. [23,27]. During transient flow conditions the microstructure is calculated as follows for both versions shown here:

$$\dot{\lambda} = k_\lambda\left(-t_{r1}\left|\dot{\gamma}_p\right|\lambda + t_{r2}\left|\dot{\gamma}_p\right|^{1/2}(1-\lambda) + (1-\lambda)\right), \tag{13}$$

where k_λ is the time constant structure build up due to Brownian motion and can only be fit to transient data when the microstructure is either being built-up or undergoing break-down. The power of (1/2) for the shear build up term is taken from the literature [8,9,12,13,20–22]. Like the EKH and IKH models above, here the $\dot{\gamma}_p$ is the plastic shear rate and used in conjunction with the evolving structure parameter. The strain and shear rate are separated into elastic and plastic components as follows

$$\dot{\gamma}_p\begin{cases}\dfrac{\dot{\gamma}}{2 - \dfrac{|\gamma_e|}{\gamma_{max}}}\dot{\gamma} \geq 0 \\ \dfrac{\dot{\gamma}}{2 + \dfrac{|\gamma_e|}{\gamma_{max}}}\dot{\gamma} < 0\end{cases} \quad \textit{and} \tag{14}$$

$$\dot{\gamma}_e = \dot{\gamma}_p - \frac{\gamma_e}{\gamma_{max}}\left|\dot{\gamma}_p\right|. \tag{15}$$

Where $\gamma_{max} = min(\gamma_0/\lambda^{1/3}, \gamma_\infty)$, and $\gamma_0 = \sigma_{y0}/G_0$ for version 1; with version 2: $\gamma_0 = \sigma_{y0}/G_R$.

Eq. (16) is the steady state of Eq. (13), and is derived by setting Eq. (13) equal to zero,

$$\lambda_{ss} = \frac{\left(t_{r2}|\dot{\gamma}|^{1/2}+1\right)}{\left(t_{r1}|\dot{\gamma}|+t_{r2}|\dot{\gamma}|^{1/2}+1\right)}, \tag{16}$$

where tr_1 is the ratio of the structural breakdown constant due to shear rate divided by the Brownian build-up constant, and tr_2 is the ratio of the shear aggregation constant to the Brownian build constant. To find a thixotropic time scale of breakage, t_{r1}, and thixotropic time scale of shear aggregation t_{r2} we set the change in shear rate with respect to time, shown in Eq. (13) equal to zero. We note here that thixotropic time scale of breakage tr_1 is the ratio of the 'constant of shear breakage' divided by the constant of Brownian build-up; and tr_2 is a ratio of the 'constant of shear build-up' divided by Brownian build-up. When fit to steady state data this will give both the thixotropic time scales t_{r1} and t_{r2}, with the third to be fit to transient data. This accounts for the time of breakdown resulting from high shear as well as the building of the fluid at lower shear rates. Our thixotropic modification to the Stickel et al. EVP only model (version *1* of our model enhancement is aa described above; while version *2* is like the SPTT-IKH construct [27]) gives a viscoelastic time scale and accounts for the time scales of thixotropic evolution, including shear break-down, shear build-up, and Brownian build-up. This is more appropriate to accurately capture the *thixo*-elasto-visco-plasticity human blood's rouleaux [12–20].

The purpose of this model is to more accurately model blood rheology in humans through the refinement the Stickel et al. EVP only, and Varchanis et al. SPTT-IKH, then use the classic IKH as a "peer-model" comparator with the least number of parameters. With the goal to more accurately correlate blood rheology to blood physiology providing a better way to correlate mechanical properties of blood to the evolving microstructure. To facilitate this process, we incorporate a more robust set of transient rheological experiments. Table 2 shows the side-by-side model comparisons with respect to equations, ODEs, and parameters for the Stickel et al. EVP-only, the IKH and both version of the proposed enhanced thixotropic EVP. The tensor for of version 1 of the ethixo-EVP is shown in Appendix A, with limited results of this version shown in the Supplemental material.

The applications of this refinement are for improved accuracy of rheological models for human blood characterization. The model could be used to identify correlations in the union of the rheology and physiology of the blood of specific patients. The ethixo EVP models will have the potential to accurately and quickly diagnose vascular disease based on differences with the physiological and rheological, mechanical fingerprint of the blood, when compared to a *'normal'* human blood rheological baseline. This may serve as beneficial to soldiers on the battlefield who require immediate transfusions with no access to proper facilities, and healthy blood is required. The performance of our proposed model's thixotropic enhancement will be assessed based on the accuracy as measured by a cost function described below. When computing these measurements, global stochastic optimization algorithms were used for determination of model parameters [8,9,29–31,35].

4. Parameter estimation

The EVP only model by Stickel et al., IKH and the enhanced ethixo-EVP models are each first fit to the steady state data of the human blood using a stochastic, global optimization algorithm with Matlab [35]. Other efforts to fit data in the literature have previously incorporated a

Table 2
Side by side model comparisons with respect to equations, ODEs, and parameters.

Stickel et al. EVP only	IKH
$\sigma_{ss} = \sigma_{y0} + k\dot{\gamma}^n$	$\sigma_{ss} = \sigma_{y0} + k\dot{\gamma}^n$
$\frac{1}{G}\dot{\sigma} + \max\left(0, \frac{\lvert\sigma\rvert - \sigma_{y0}}{k\lvert\sigma\rvert^n}\right)^{1/n}\sigma = \dot{\gamma}$	$\gamma_{tot} = \gamma_e + \gamma_p \leftrightarrow \dot{\gamma}_{tot} = \dot{\gamma}_e + \dot{\gamma}_p$ $\dot{\gamma}_p = \mathbf{n}^p\lvert\dot{\gamma}_p\rvert;\ \mathbf{n}^p = \frac{\sigma - \sigma_{back}}{\lvert\sigma - \sigma_{back}\rvert}$ $\lvert\dot{\gamma}_p\rvert = \left(\frac{\lvert\sigma - \sigma_{back}\rvert}{k}\right)^{1/n}$ $\dot{A} = \dot{\gamma}_p - qA\lvert\dot{\gamma}_p\rvert$ $\sigma = G\gamma_e \qquad \sigma_{y0} = C/q$
ODEs: 1; **Parameters(4)**: k, n, G, σ_{y0}	**ODEs: 3; Algebraic: 3** **Parameters(6):** k, n, G, C, q, σ_{y0}
ethixo-EVP (vers. 1 & 2)	
$\sigma_{ss} = \lambda\sigma_{y0} + \lambda\eta_{ST}\dot{\gamma} + \eta_\infty\dot{\gamma} \qquad \dot{\lambda} = k_\lambda\left(-tr_1\lvert\dot{\gamma}_p\rvert\lambda + tr_2\lvert\dot{\gamma}_p\rvert^{1/2}(1-\lambda) + (1-\lambda)\right)$ $\lambda_{ss} = \frac{(tr_2\lvert\dot{\gamma}\rvert^{1/2}+1)}{(tr_1\lvert\dot{\gamma}\rvert + tr_2\lvert\dot{\gamma}\rvert^{1/2}+1)} \qquad \gamma_{tot} = \gamma_e + \gamma_p \leftrightarrow \dot{\gamma}_{tot} = \dot{\gamma}_e + \dot{\gamma}_p$	
$vers.1: \frac{1}{G}\dot{\sigma}_{tot} + \max\left(0, \frac{\lvert\sigma_{tot}\rvert - \lambda\sigma_{y0}}{(\lambda\eta_{ST} + \eta_\infty)\lvert\sigma_{tot}\rvert}\right)\sigma_{tot} = \dot{\gamma}$ $\gamma_{max(1)} = \min\left(\gamma_{0(1)}/\lambda^{1/3}, \gamma_\infty\right), \gamma_{0(1)} = \sigma_{y0}/G$ $vers.2: \frac{1}{G_R}\dot{\sigma}_p + \max\left(0, \frac{\lvert\sigma_s\rvert - \lambda\sigma_{y0}}{(\lambda\eta_{ST})\lvert\sigma_s\rvert}\right)\sigma_s = \dot{\gamma}_p$ $\sigma_{tot} = \sigma_s + \eta_\infty\dot{\gamma}$ $\gamma_{max(2)} = \min\left(\gamma_{0(2)}/\lambda^{1/3}, \gamma_\infty\right), \gamma_{0(2)} = \sigma_{y0}/G_R$	$\dot{\gamma}_p \begin{cases} \frac{\dot{\gamma}}{2 - \frac{\lvert\gamma_e\rvert}{\gamma_{max}}} & \dot{\gamma} \geq 0 \\ \frac{\dot{\gamma}}{2 + \frac{\lvert\gamma_e\rvert}{\gamma_{max}}} & \dot{\gamma} < 0 \end{cases}$ $\dot{\gamma}_e = \dot{\gamma}_p - \frac{\gamma_e}{\gamma_{max}}\lvert\dot{\gamma}_p\rvert$
ODEs: 3; Algebraic: 2; **Parameters(8)**: η_{st}, η_∞, tr_1, tr_2, k_λ, G, σ_{y0}, γ_∞	

weighting function in terms of uncertainty of the data. More details on this parameter fitting algorithm can be found here [8,9,22,35]. The steady state residual sum of squares is shown below, note that the difference between model prediction of the stress and the actual stress is normalized by the actual stress since we are fitting stress data over several orders of magnitude (for steady state only). The cost function which is minimized for the steady state parameter fit is defined as follows

$$F_{cost,ss} = \sqrt{\sum_{i=1}^{m}\left(\frac{(y_i - f_i)}{y_i}\right)^2} \Big/ m, \tag{17}$$

where y_i is the steady state measurement of stress, f_i is the model prediction and m are the number of steady state data points *(m=16 for the steady state fits)*. The steady state data is normalized with stress measurement since typically with a steady state flow curve, there is shear rate data over several decades. With the steady state parameters then fixed, the single transient parameter for the EVP only/IKH models, G, and the two transient parameters for the enhanced ethixo-EVP models, G (or G_R) and k_λ, are then fit to a sequence of three step-up in shear rate, and three step-down in shear rate experiments simultaneously. This step is accomplished using the ode23s ordinary differential equation solver in the Matlab toolbox. The differences between model prediction and stress are not normalized during the transient parameter fitting

procedure, and therefore have units of (Pa). The residuals are accumulated and minimized with the optimization algorithm over log spaced points throughout the step-up and step-down rheological experiments [8,9]. The cost function that is used for the transient parameter fit is thusly defined

$$F_{cost,trans} = \sum_{j=1}^{6}\left(\sqrt{\sum_{i=1}^{m}(y_i - f_i)^2}\Big/ m\right)\Big/ 6. \tag{18}$$

Eqs. (17) and (18) are minimized, and optimum values for model parameters determined through a stochastic algorithm and the parallel tempering algorithm use is detailed here *(m=65 for the step up/down in shear rate fits)* [32]. Our parallel tempering algorithm maintains *10* number of Boltzmann energy levels, whereby the higher the energy level of the parallel run, the more freedom to explore the parameter space is allowed. At predetermined algorithm 'pause' points, nearest neighbor energy levels compare cost function values. The set of model parameters with the lowest cost function values are passed to the *"lowest"* Boltzmann energy level. Over the run of the algorithm the best fit parameters naturally wind up in the lowest energy level [35]. Lastly, a comparison will be made to determine if the addition of new parameters contributed to a better representation of the actual stress signatures by incorporating straight cost function comparison of the steady state and transient data, as well as use of the Akaike Information Criteria (AIC), and Bayesian Information Criteria (BIC) [35,36] shown below which assesses a penalty for additional model parameterization

$$AIC = 2K + 2\ln(RSS), \tag{19}$$

$$BIC = 2Kln(N) + 2ln(RSS), \tag{20}$$

where

$$RSS = \sum_{i=1}^{m=16}\left(\frac{(y_i - f_i)}{y_i}\right)^2, \tag{21}$$

where K is number of model parameters, N is number of data points and RSS is the residual sum of squares [35,36]. Eq. (21) shows the RSS for the case of the steady state residual sum of squares [29,30,35,36]. We note here that the AIC is in fact an approximation of the full Bayesian inference integrals, and not considered here because the computation is too costly. Much relevant work has been done with the full Bayesian inference approach by Freund and Ewoldt [37], however for our purposes the penalty for additional parameters is appropriate.

The results section will show the fits, and numerically compare the cost functions and Akaike Information Criteria (AIC). The results will also show a comparison of the models in predicting a sequence of transient, linear, and nonlinear flows graphically and numerically with F_{cost}. using one set of human blood rheology data from Donor 3. All the donor results are shown in the Supplemental Material.

Once the model parameters have been fit to the steady state and a sequence of step up and step down in shear rate data, the parameters are fixed, and then utilized to predict a sequence of transient rheological experiments, including small amplitude oscillatory shear amplitude sweep, a sequence of triangle ramp experiments, LAOS, and unidirectional oscillatory shear flow (UD-LAOS) [7,8,9,12,21,22]. The cost functions will be accumulated for each sequence of experiments and compared for the ethixo-EVP models, IKH and the EVP only model for four donors. The results will be shown graphically and numerically. Recall for oscillatory shear flow starting with an applied strain of γ =

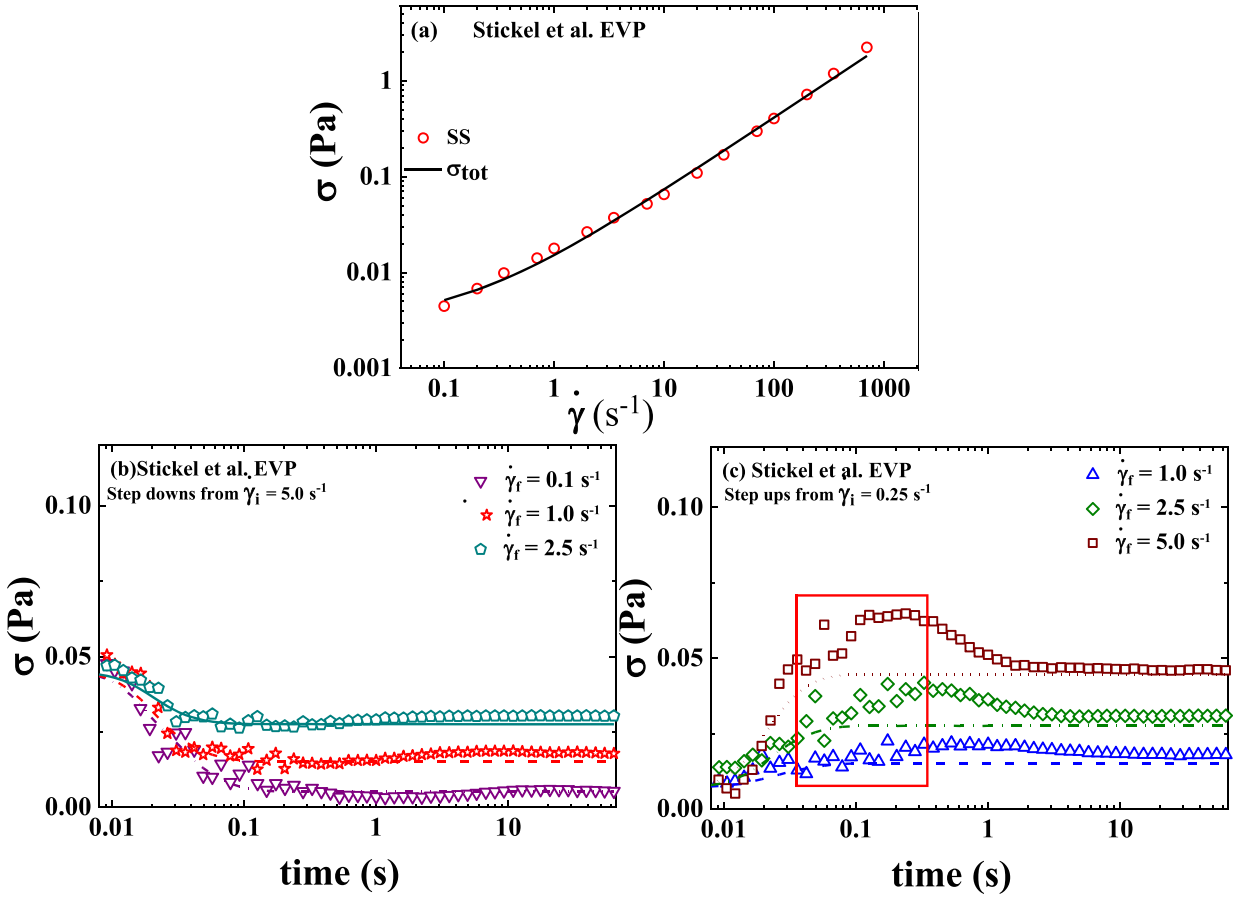


Fig. 1. Stickel et al. EVP only model (a) Steady state fit; (b) Step downs from a shear rate of $\dot{\gamma}_i$ =5s^{-1} to 0.1, 1.0 and 2.5s^{-1}; (c) Step ups from a shear rate of 0.25s^{-1} to 1.0, 2.5 and 5.0s^{-1} (points represent data; lines represent model fit). The red box is used here to highlight the weakness in the model to accurately fit the data. (Donor 3) [42].

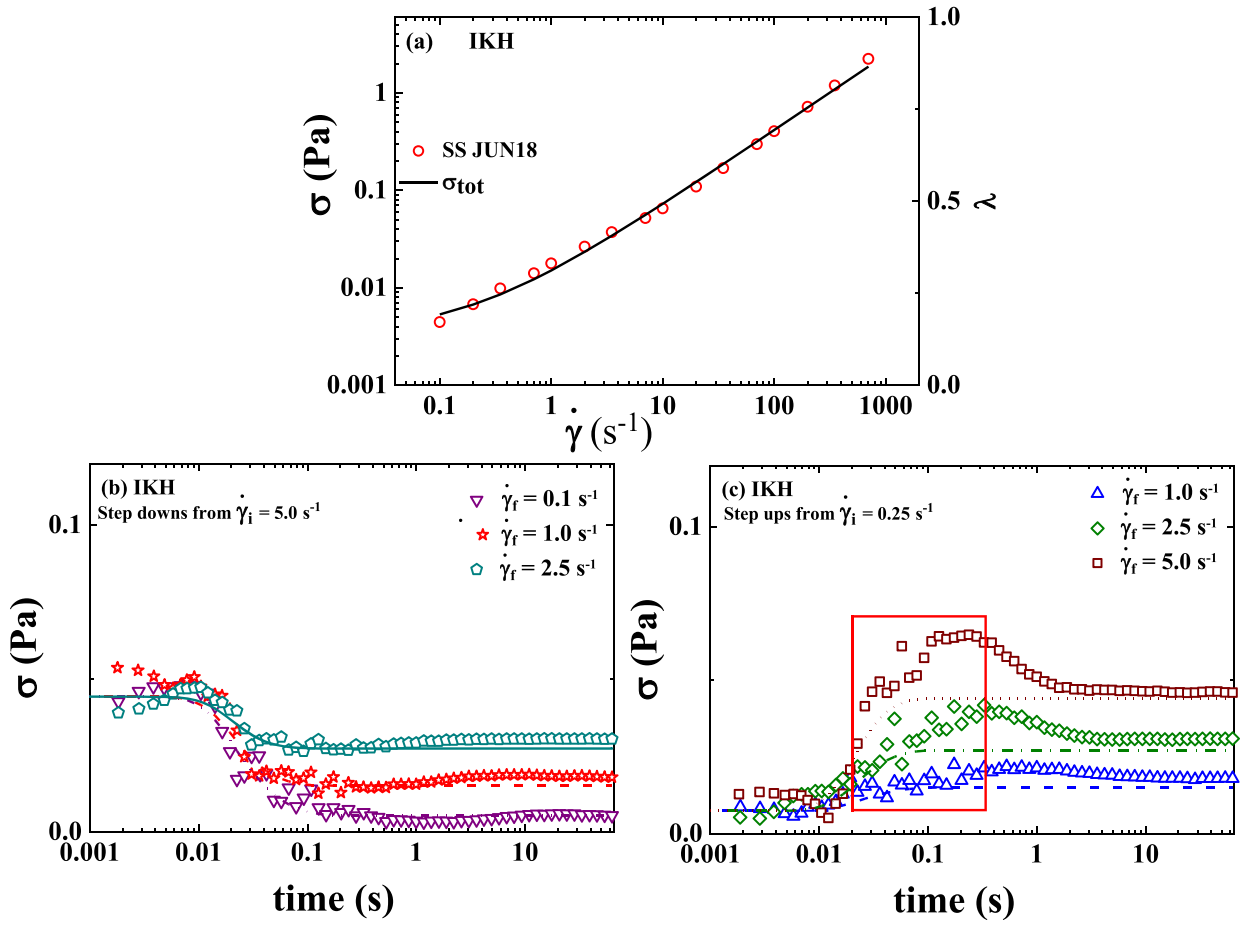


Fig. 2. IKH model (a) Steady state fit; (b) Step downs from a shear rate of$\dot{\gamma}_i$ =5s^{-1} to 0.1, 1.0 and 2.5s^{-1}; (c) Step ups from a shear rate of 0.25s^{-1} to 1.0, 2.5 and 5.0s^{-1} (points represent data; lines represent model fit). The red box is used here to highlight the weakness in the model to accurately fit the data. (Donor 3) [42].

$\gamma_0 sin(\omega t)$, with the first time derivative the shear rate $\dot{\gamma} = \omega\gamma_0 cos(\omega t)$, where γ_0 is the strain amplitude and ω is the oscillation frequency in (radians/s). For SAOS this is the linear regime whereby the stress signal is consisting of a liquid like response described by G'' (Pa) and a solid like response described by G' (Pa). Additionally, we note that the definition of SAOS flow implies the "microstructure" remains intact. It is noted here that G'' is related to viscosity as follows: $\eta' = \frac{G''}{\omega}$ and $\eta'' = \frac{G'}{\omega}$. We note here that both G' and G'' can be utilized to calculate the real and imaginary parts of the complex viscosity by dividing by the angular frequency, ω. Eq. (22) shows the stress reconstruction in the linear regime

$$\sigma(t) = \gamma_0(G' \sin(\omega t) + G''\cos(\omega t)), \tag{22}$$

where G' is the storage modulus and G'' is the loss modulus. As nonlinearity is approached, the full reconstructions using a Fourier decomposition of the stress signal over a cycle at steady alternance given by

$$\sigma(t) = \gamma_0 \sum_{j=1,odd}^{N} \left[G'_j \sin(j\omega t) + G''_j \cos(j\omega t)\right]. \tag{23}$$

Where again, y_0 is the strain amplitude, ω is the frequency of oscillation, j is the harmonic number, G' is the storage modulus (when j=*1*), and G'' is the loss modulus (when j=*1*), and the G'_n and G''_n ($j > 1$) terms are referred to as the higher order moduli. If G' and G'' together, can describe the stress signal alone, one is in the linear regime, which is now also known by its retronym 'small amplitude oscillatory shear' (SAOS). Upon transition into the regime where a single harmonic can no longer accurately describe and/or recreate the oscillatory stress signal is the MAOS and then LAOS regime. In approaching the LAOS regime one can define $I_{3/1}$ as the intensity of the 3rd harmonic divided by the intensity of the 1st harmonic as

$$I_{3/1} = \frac{\sqrt{G_3'^2 + G_3''^2}}{\sqrt{G_1'^2 + G_1''^2}}, \tag{24}$$

with the $G'_{jj}{}''$ values from the first and third harmonic. The LAOS predictions shown in the results section below compare LAOS sequence of strain amplitudes, over three varying frequencies, starting at time t=0s, with cost function accumulated over the three cycles as the difference between the model prediction and the data. The triangle ramp experiments consist of a ramp up of shear rate starting at time t=0, to a maximum shear rate, followed by a ramp down back to zero as shown in Eq. (25)

$$\begin{aligned} \dot{\gamma} &= \alpha t \quad t \le t_{1/2} \\ \dot{\gamma} &= \alpha(t_{max} - t) \quad t > t_{1/2} \end{aligned}, \tag{25}$$

where t_{max} is the maximum time of the triangle ramp, $t_{1/2}$ is the $t_{max}/2$, and α is the slope of the increase in shear rate [7,31].

Uni-direction large amplitude oscillatory shear flow is a linear superposition between a steady state shear rate and an oscillating flow with the same magnitude strain amplitude as the steady state shear value. Eqs. (26) and (27) show the uni-directional strain and shear rate respectively where again ω is frequency of oscillation and γ_0 is the strain amplitude [8,9,21,22,32]

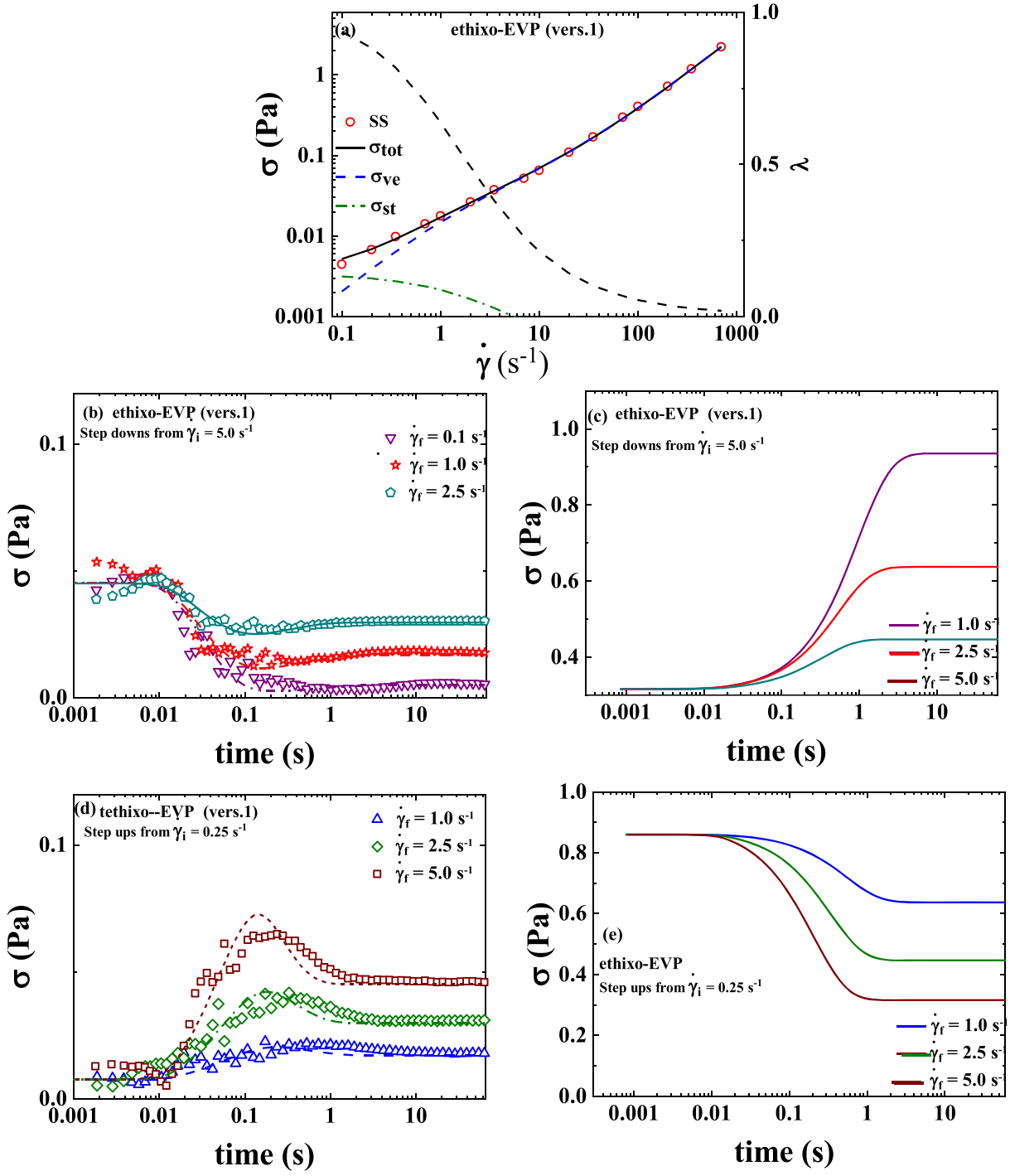


Fig. 3. Enhanced thixotropic EVP (vers.1) model (a) Steady state fit; (b) Step downs from a shear rate of$\dot{\gamma}_i$ =5s^{-1} to 0.1, 1.0 and 2.5s^{-1}; (c) structure parameter during step downs; (d) Step ups from a shear rate of $\dot{\gamma}_i$ = 0.25s^{-1} to 1.0, 2.5 and 5.0s^{-1};(e) structure parameter during steps ups (points represent data; lines represent model fit in b, d; lines represent structure parameter prediction in c, e). (Donor 3) [42].

$$\gamma = \gamma_0 \sin(\omega t) + \gamma_0 \omega t, \tag{26}$$

$$\dot{\gamma} = \gamma_0 \omega \cos(\omega t) + \gamma_0 \omega. \tag{27}$$

5. Results: model *fitting* and *predictions;* and sequence of physical processes correlations

5.1. Steady state and stepwise changes in shear rate fitting

Figs. 1–4a show similar ability to predict the steady state flow; however, there is significant loss of ability to show the overshoot in the step up in shear rate tests shown in Figs. 1–4 and highlighted with a red box. The EVP and IKH models miss the overshoot whereby the ethixo EVP models can both qualitatively capture the rheological behavior more accurately. This is due to the ability of the *ethixo*-EVP models to simultaneously incorporate a viscoelastic time scale and a set of appropriate thixotropic timescales in real time. Table 3 shows the best fit model parameters of both the enhanced *thixo*-EVP models, the IKH and the EVP only model, respectively. Recalling, that both models were fit with the same procedure by first fitting the steady state parameters to

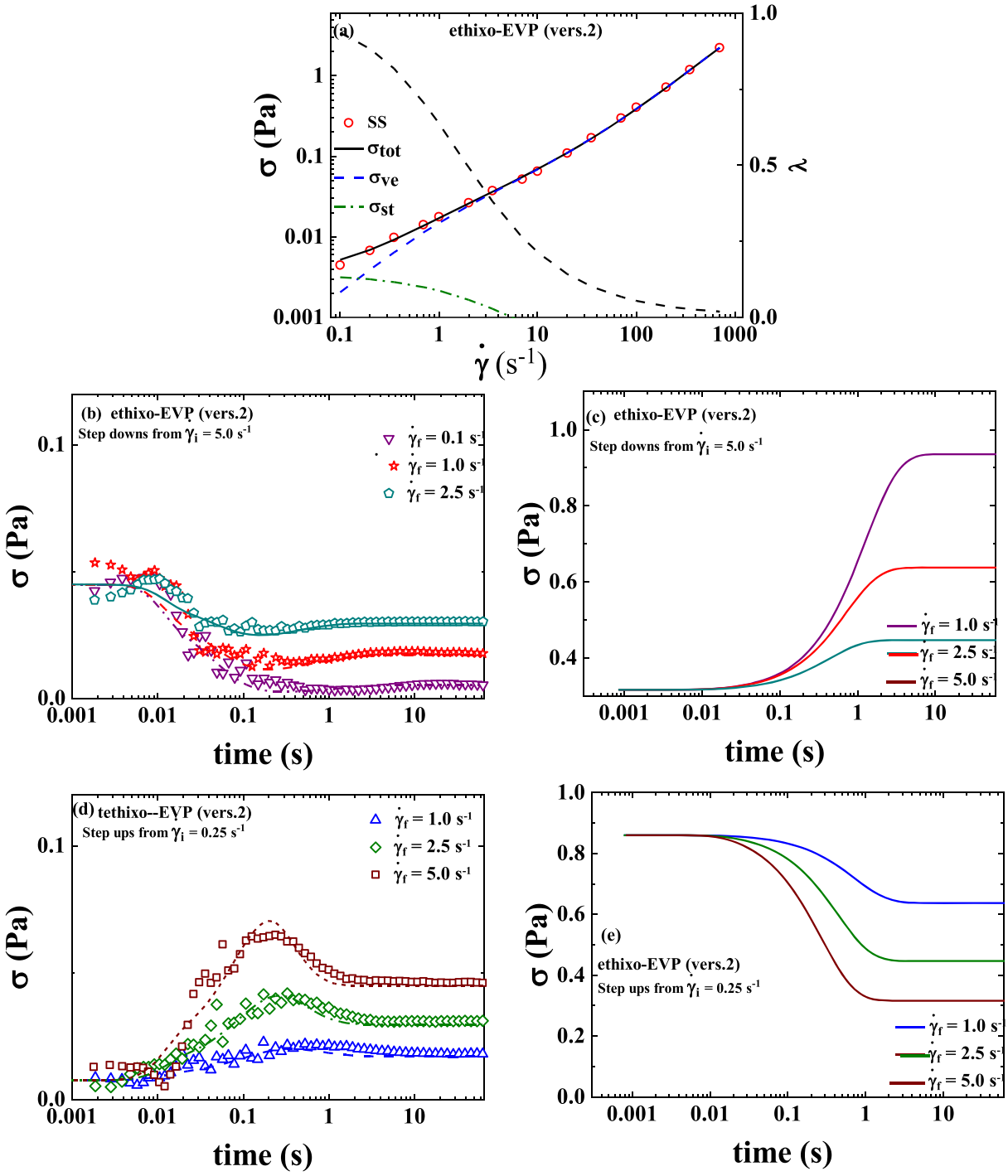


Fig. 4. Enhanced thixotropic EVP (vers.2) model (a) Steady state fit; (b) Step downs from a shear rate of$\dot{\gamma}_i$ =5s^{-1} to 0.1, 1.0 and 2.5s^{-1}; (c) structure parameter during step downs; (d) Step ups from a shear rate of $\dot{\gamma}_i$ = 0.25s^{-1} to 1.0, 2.5 and 5.0s^{-1};(e) structure parameter during steps ups (points represent data; lines represent model fit in b, d; lines represent structure parameter prediction in c, e) (Donor 3) [42].

the steady state data; followed by a fit to three sets of step-up in shear rate, and three sets of step-down in shear rate tests to fit the transient parameters. Table 4 shows a comparison of the F_{cost} for steady state and transient model fits for Donor 3.

Table 4 shows the normalized cost function of the steady state parameter fit, as well as the residual sum of squares along with AIC and BIC for the steady state fit, and the transient step up/down in shear rate fits. Table 4 shows the value of F_{cost} and *RSS* decreasing, which is expected by adding additional parameters to capture the physics of rouleaux formation and breakdown, and analogous contributions to total stress, however this has come with the cost of an increase in *AIC*. This is the trade-off of the additional parameters added for robust thixotropy modeling and more accurate predictions of total stress, and part of logic to not incorporate the SPTT-IKH and ML-IKH (due to large numbers of parameters). Those modeling TEVP materials must perform a simple "cost-benefit" analysis with respect to model fitting to rheological data, and model predictions of total stress. In the case of human blood modeling, we have assumed risk of additional model parameters

Table 3
Best fit model parameters of enhanced thixotropy-EVP, IKH, and Stickel et al. EVP.

ethixo-EVP (vers. 1) Par.		ethixo-EVP (vers. 2) Par.		IKH Par.		Stickel et al. EVP only Par.	
σ_{y0} (Pa)	0.0034	σ_{y0} (Pa)	0.0034	σ_{y0} (Pa)	0.0034	σ_{y0} (Pa)	0.0034
η_∞ (Pa s)	0.0028	η_∞ (Pa s)	0.0028	k (Pa s^n)	0.0117	k (Pa s^n)	0.0117
η_{ST} (Pa s)	0.019	η_{ST} (Pa s)	0.019	n	0.774	n	0.774
G (Pa)	0.29	G_R (Pa)	0.14	G (Pa)	0.448	G (Pa)	0.45
tr_1 (s)	0.77	tr_1 (s)	0.77	C	0.0028		
tr_2 ($s^{1/2}$)	0.34	tr_2 ($s^{1/2}$)	0.34	q (Pa^{-1})	0.926		
k_λ (s^{-1})	0.88	k_λ (s^{-1})	0.67				
γ_∞	1	γ_∞	1				

Table 4
F_{cost}, RSS, AIC, BIC values for steady state and step up/down experiment fits using enhanced thixo-EVP, IKH, and Stickel et al. EVP models. (F_{cost} (SS) is non-dimensional; F_{cost} (trans) has units of Pa) (Donor 3).

Donor3	ethixo-EVP (vers. 1)	ethixo-EVP (vers. 2)	IKH	Stickel et al EVP only
F_{cost} (SS)	0.014	0.014	0.028	0.028
n (points)	16	16	16	16
RSS (SS)	0.051	0.051	0.199	0.199
k (par.)	5	5	3	3
AIC (SS)	4.04	4.04	2.77	2.77
BIC (SS)	21.77	21.77	13.41	13.41
F_{cost} (trans.)	0.00041	0.00045	0.00061	0.00061
n (points)	390	390	390	390
RSS (trans.)	0.026	0.031	0.0573788	0.057
N (par.)	8	8	5	4
AIC (trans.)	8.71	9.02	4.28	2.29
BIC (trans)	76.37	88.48	53.95	42.02

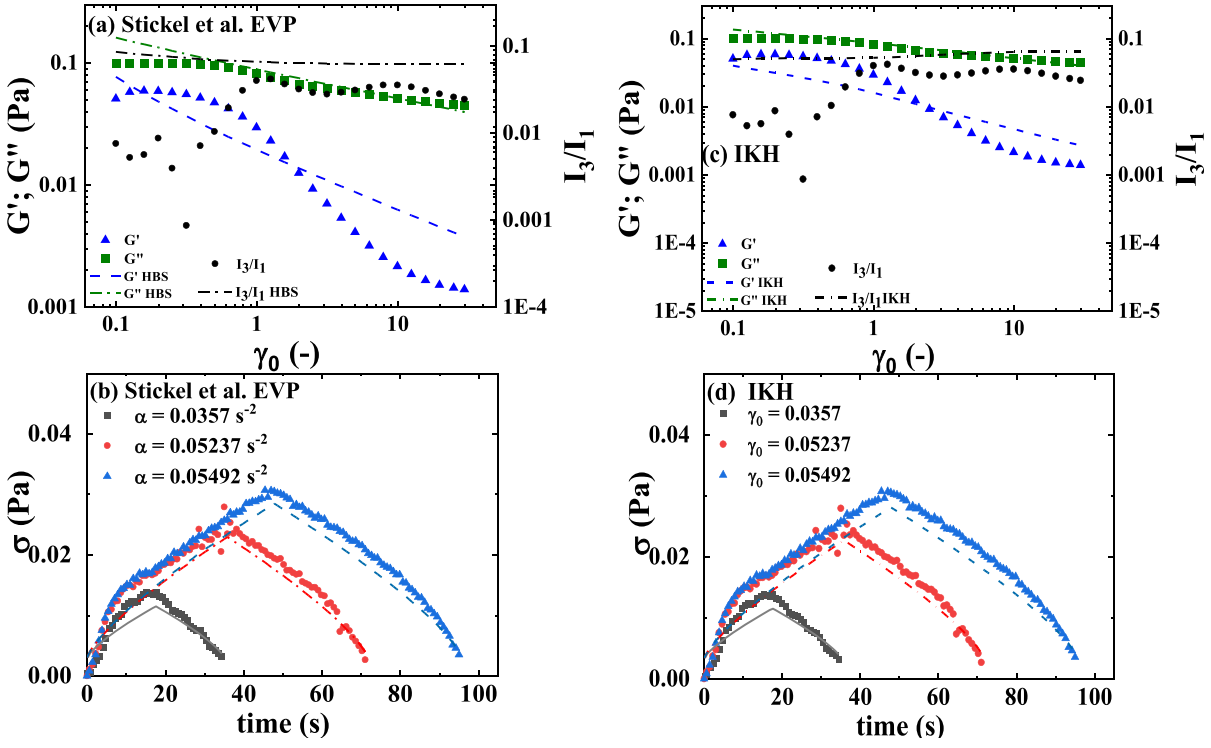


Fig. 5. (a) Stickel et al. EVP only model amplitude sweep (points) at $\omega = 12.56(rad/s)$ and model predictions (lines); (b) Stickel et al. EVP model triangle ramp predictions with three α values listed in figure; (c) IKH model amplitude sweep (points) at $\omega = 12.56(rad/s)$ and model predictions; (d) IKH model triangle ramps predictions, with three α values listed in figure (Donor 3) [42].

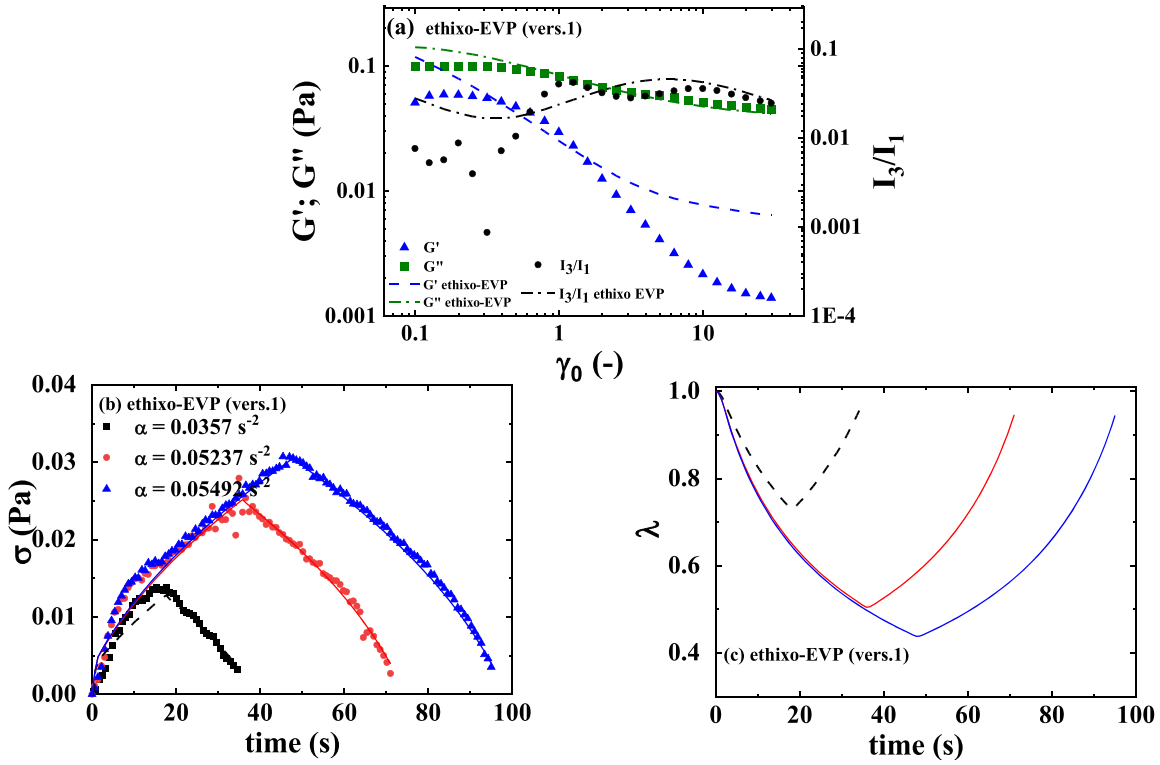


Fig. 6. (a) Enhanced thixotropic EVP (vers.1) model amplitude sweep, $\omega = 12.56(rad/s)$ (points) model predictions (lines); (b) enhanced thixotropy-EVP model triangle ramp predictions; and (c) ethixo-EVP (vers.1) model structure parameter predictions (Donor 3) [42].

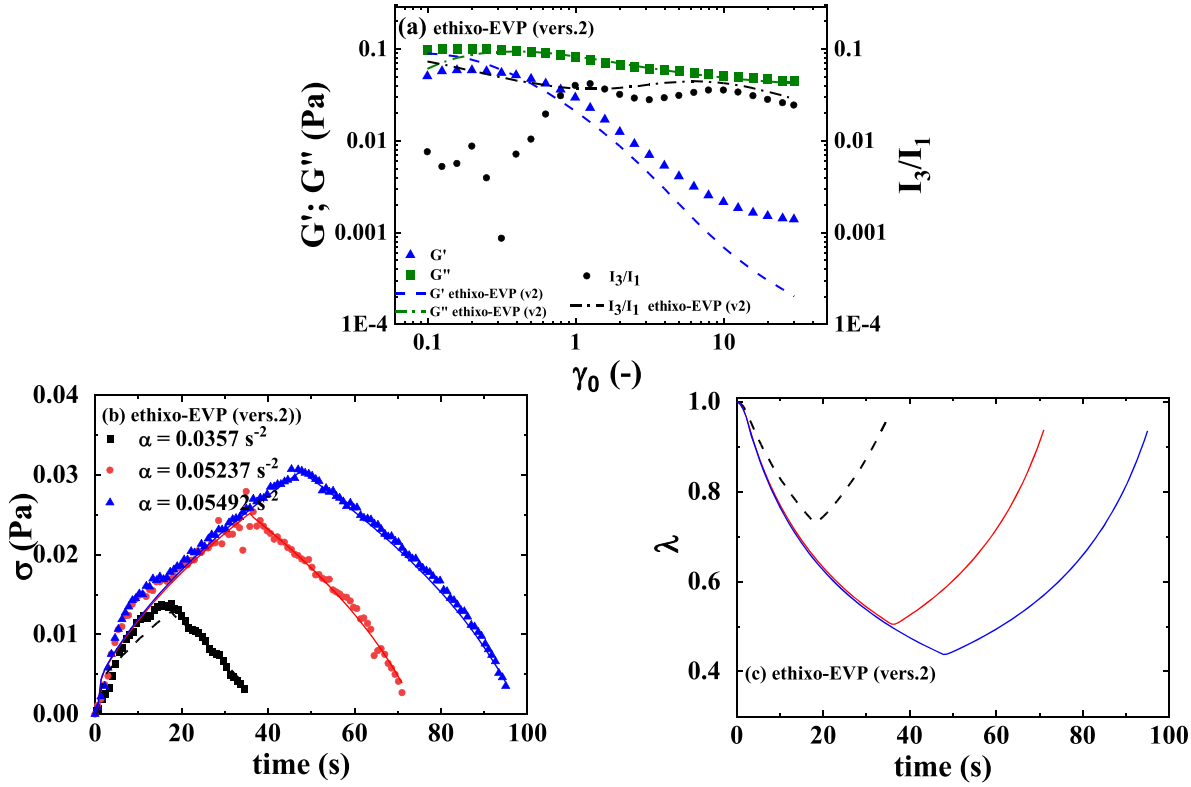


Fig. 7. (a) Enhanced thixotropic EVP (vers.2) model amplitude sweep, $\omega = 12.56(rad/s)$ (points) model predictions (lines); (b) enhanced thixotropy-EVP model triangle ramp predictions; and (c) ethixo-EVP (vers.2) model structure parameter predictions (Donor 3) [42].

Table 5
Comparison of F_{cost} for thixo-EVP, IKH and EVP only using Donor 3 triangle ramp, and SAOS predictions.

	ethixo EVP (vers.1)	ethixo EVP (vers.2)	IKH	Stickel et al EVP only
Ramps:	**F_{cost}**			
Ramp1 (Pa)	1.38E-04	1.34E-04	2.58E-04	2.27E-04
Ramp2 (Pa)	2.06E-04	1.92E-04	2.97E-04	3.53E-04
Ramp3 (Pa)	1.05E-04	1.02E-04	2.52E-04	1.94E-04
Ave. $F_{cost,ramps}$(Pa)	**1.50E-04**	**1.43E-04**	**2.69E-04**	**2.58E-04**
SAOS	**F_{cost}**			
Amp. Sweep (Pa)	0.0024	0.0016	0.0022	0.0019
I_3/I_1	0.0022	0.0058	0.0094	0.0068

since capturing the flow induced build-up and breakdown of rouleaux is critical to understanding the physics of human blood flow and making further predictions of pathological blood and its relationship to health. By incorporating the lower parameter thixotropic framework to the EVP only model there is a cost of two more parameters (as the powers of shear breakage and shear buildup are fixed). The benefit is seen directly in comparing Figs. 1–4. When fitting and making predictions with rheological materials containing microstructure it is imperative to: 1. Choose a rheological model that allows prediction of thixotropy and has a thixotropic term (aka. microstructure); and 2. That the thixotropic portion of the model adequately has appropriate mathematics to model the physics of evolution of microstructure, which necessarily requires a shear breakage term, and Brownian rebuilding term, and a shear aggregation term [12,13,17,19,21,22]. This will give the thixotropic portion of the model all the capability to robustly model the thixotropic portion of the various rheological responses accurately under all flow conditions including small amplitude oscillatory shear, large amplitude oscillatory shear, uni-directional oscillatory shear, triangle ramp experiments and cessation of flow.

5.2. Comparison of models using SAOS (amplitude sweep) and triangle ramp predictions

In this section we compare predictions with both models, by holding the best fit parameter values from above constant. We compare an amplitude sweep at frequency of 12.56 (rad/s) and three separate triangle ramp experiments. To calculate the F_{cost} for the amplitude sweep

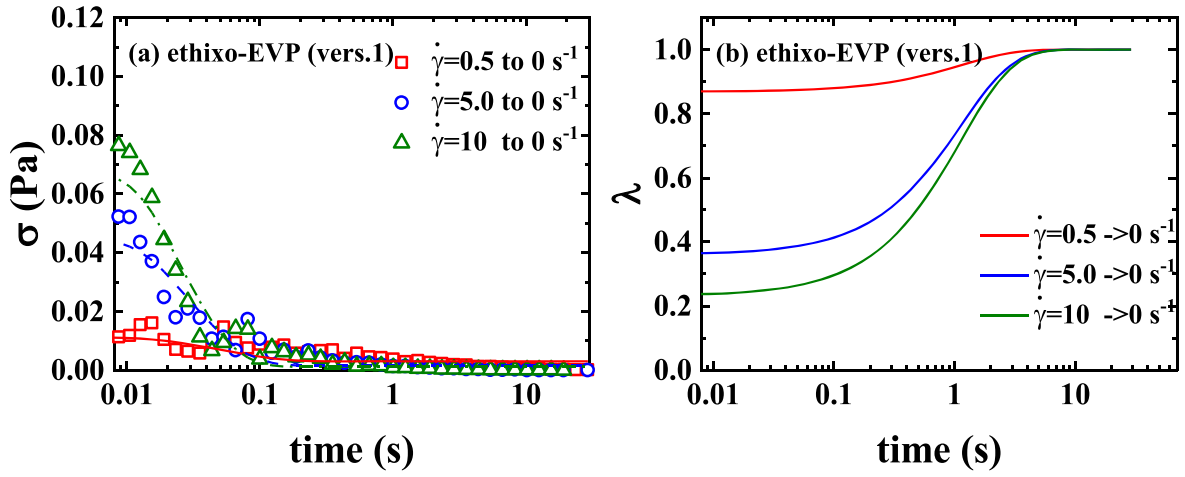


Fig. 8. (a) Cessation of flow data (points) and predictions (lines) and (b) structure parameter predictions during cessation of flow experiments using enhanced thixotropic EVP (vers.1) model Donor 3 [42].

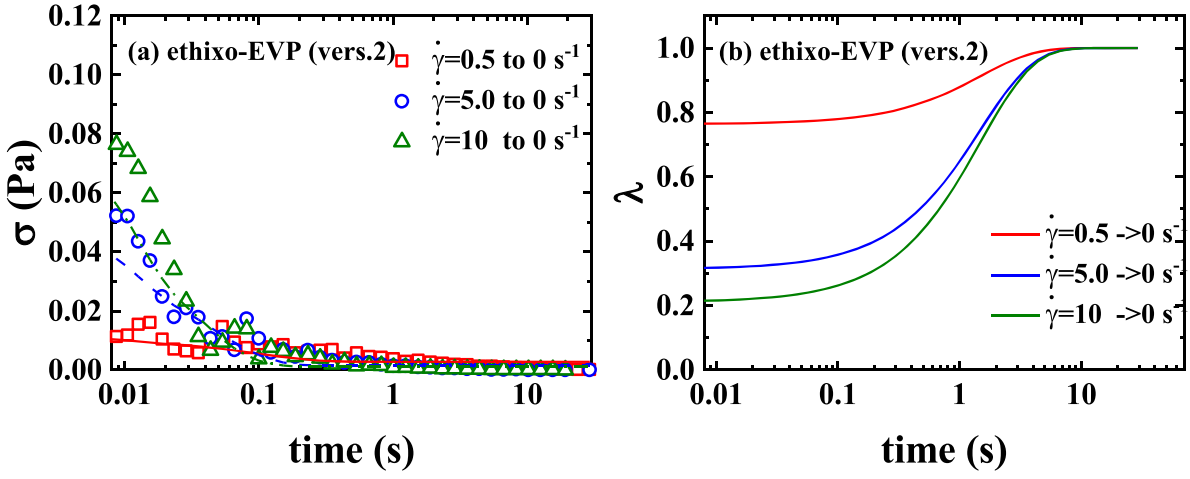


Fig. 9. (a) Cessation of flow data (points) and predictions (lines) and (b) structure parameter predictions during cessation of flow experiments using enhanced thixotropic EVP (vers.2) model Donor 3 [42].

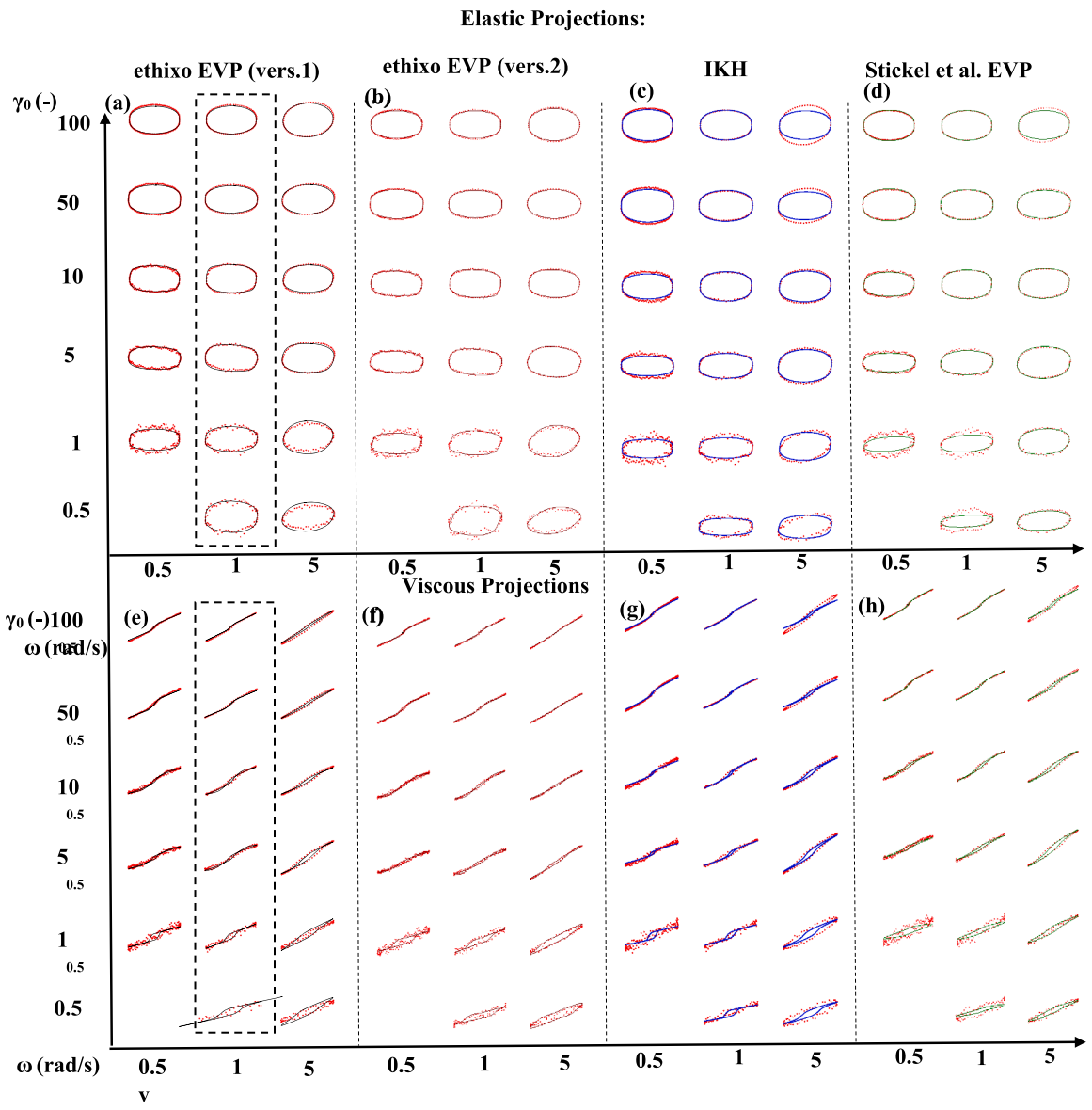


Fig. 10. (a-d) Elastic LAOS projections of ethixo EVP vers.1 (black lines), ethixo EVP vers.2 (maroon lines); IKH (green lines) and Stickel et al. EVP (blue lines); (e-h) viscous LAOS projections of ethixo EVP (black lines), ethixo EVP vers.2 (maroon lines), IKH (green lines) and Stickel et al. EVP (blue lines). Red points are data; x-axis is frequency (rad/s), y-axis is strain amplitude (-) (Donor 3) [42].

experiment we run the respective models until alternance is reached; and with the last period the quantities $G^{'}, G^{''}, G_3^{'} and G_3^{''}$ are evaluated through harmonic regression [8,9]. This is executed with Matlab by solving the linear equation $\mathbf{Ax} = \mathbf{b}$ where $\mathbf{A}$ is the array containing the transient harmonic data (only 1st and 3rd harmonics included), $\mathbf{b}$ is a vector containing the transient measures, and $\mathbf{x}$ is the vector containing the moduli for each harmonic [8,9]. Once the respective sets of model parameters are fixed after the transient parameter fit using parallel tempering, they are held constant. During the prediction phase we first compare transient predictions to three separate triangle ramp experiments.

Figs. 5,6a and b show qualitatively the fits using in Fig. 5a EVP only and Fig. 5c the IKH. Figs, 6 and 7 show both versions of the ethixo-EVP. Note that Figs. 6c and 7c show the evolving microstructure of the human blood, which the EVP only and IKH model do not. Table 5 below shows a comparison in F_{cost} for each of the four predictions side by side. Once again, the ethixo-EVP models can more successfully capture the thixo-elasto-visco-plastic dynamic behavior by bringing in simultaneously a viscoelastic timescale and appropriate thixotropic timescales. The model contains additional parameters, however because human blood modeling requires a thixotropic term to adequately account for formation and break down of rouleaux, the parameters are thus justified. The added insight to the microstructure of human blood during these transient flows is insightful and worth the cost of the additional parameters. We have thusly shown that the enhanced *thixo*-EVP models are better able to predict the complex rheology of human blood, at the cost of additional parameters. Donor 1-4 fitting results, and F_{cost} results are shown for EVP only, IKH and both ethixo-EVP models in the Supplemental Material. The Supplemental Material also shows the best fit parameters, with limited fits show for Donor 4.

5.3Validation. of enhanced thixotropy EVP models using cessation of flow, large amplitude oscillatory shear (LAOS) flow, and uni-directional LAOS

With the *ethixo*-EVP model's enhanced ability to predict dynamic rheological data we present in this section predictions using the model with cessation of flow, LAOS, and UD-LAOS experiments. Figs. 8a and 9a

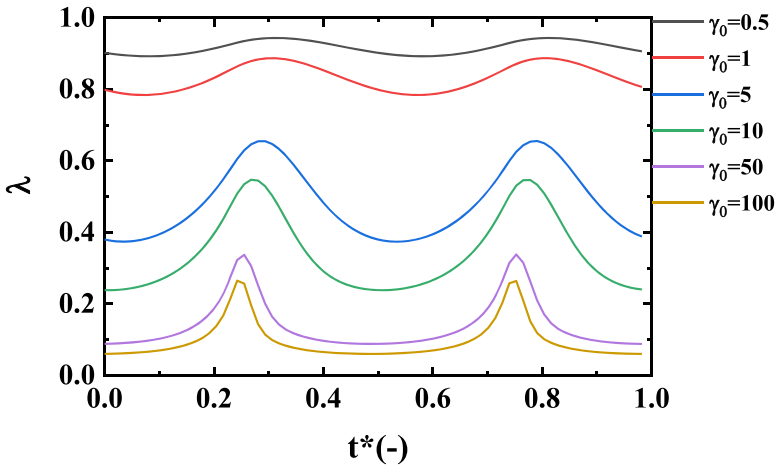


Fig. 11. Structure parameter prediction using thixo-EVP model vs. non-dimensional time (period length used to normalize time) of amplitude sweep of LAOS at $\omega = 1$ (rad/s). (Structure parameter of analogous Lissajous-Bowditch plots from Fig. 7 with a black, dashed box, shown here.)

Table 6
LAOS F_{cost} comparison (Donor 3, over all LAOS experiments ω=0.5–5 (rad/s), γ_0=0.5 100(1).

		ethixo EVP (vers. 1)	ethixo EVP (vers. 2)	IKH	Stickel et al EVP only
ω=0.5 (rad/s)	**γ_0=1(-)**	0.00026	0.00022	0.00033	0.00029
	γ_0=5	0.00023	0.00019	0.00032	0.00044
	γ_0=10	0.00038	0.00022	0.00029	0.00055
	γ_0=50	0.00055	0.00065	0.00041	0.00102
	γ_0=100	0.00092	0.00122	0.00068	0.00147
		0.00047	**0.00050**	**0.00041**	**0.00075**
ω=1.0 (rad/s)	**γ_0=0.5 (-)**	0.00031	0.00027	0.00046	0.00028
	γ_0=1	0.00029	0.00035	0.00054	0.00037
	γ_0=5	0.00035	0.00040	0.00038	0.00046
	γ_0=10	0.00067	0.00052	0.00038	0.00051
	γ_0=50	0.00127	0.00134	0.00098	0.00108
	γ_0=100	0.00215	0.00225	0.00090	0.00093
		0.00084	**0.00086**	**0.00061**	**0.00060**
ω=5.0 (rad/s)	**γ_0=0.5 (-)**	0.00071	0.00064	0.00036	0.00094
	γ_0=1	0.00075	0.00078	0.00041	0.00114
	γ_0=5	0.00113	0.00057	0.00109	0.00163
	γ_0=10	0.00186	0.00133	0.00153	0.00246
	γ_0=50	0.00659	0.00495	0.00779	0.01750
	γ_0=100	0.01271	0.00872	0.02549	0.04668
		0.00396	**0.00283**	**0.00611**	**0.01172**
Ave $F_{cost,LAOS}$ (Pa)		**0.00176**	**0.00139**	**0.00238**	**0.00436**

show the data and predictions with *ethixo*-EVP models of three cessation of flow experiments from 0.5s-1, 5.0s-1 and 10s- to zero. We acknowledge that there are experimental limitations with respect to measuring the stress on the ARESG2 at short times, and great care must be taken to compare model responses here. Qualitatively the enhanced thixotropy EVP model can capture the dynamics of the cessation properly, and using Figs. 8b and 9b as a guide, one can estimate the time human blood requires to fully restructure after cessation of flow to be approximately 3.5-5 seconds. This time scale is on the order the interval between heartbeats in the average human being.

Fig. 10a–d below shows the elastic projections of the data and all model predictions, while Fig. 10e–h shows the viscous projections of the data with all model predictions for a frequency range 0.5 to 5 (rad/s) and strain amplitude range 0.5 – 100. The LAOS data indicate that as the shear rate increases the material becomes more and more viscous or "liquid-like". The smaller the shear rate the more elastic or "solid-like" behavior is seen. We note here that the LAOS data presented in Fig. 10 has noise present in the data at the lower strain amplitudes. The ethixo-EVP models also predict crossovers some of the viscous projections. The shear rate of the rheometer, as well as the changing flow direction two times per period causes the microstructure to mostly be broken, more so at the higher strain amplitudes. The corresponding values of structure parameter for the $\omega = 1(rad/s)$ are shown vs. normalized time in Fig. 11, indicating that, as shown in Fig. 10 the larger the strain amplitude the less structured the blood is. The F_{cost} of each model's prediction of each LAOS experiment, along with average values of F_{cost} is shown in Table 6. The ethixo EVP models are both able to capture the more nonlinear features of LAOS with the addition of thixotropy.

Analysis of the elastic projection data (red circles) in Fig. 10a-d shows that the stress vs. strain indicates the signal is predominantly a viscous signal. That the human blood is more "liquid-like" than it is "solid-like at all of the frequency $(\omega = 0.5, 1, 5rad/s)$ and strain amplitude $(\gamma_0 = 0.5, 1, 5, 10, 50, \text{and} 100(-))$ combinations. This is because the shape of the elastic projections is predominantly elliptical/ circular. Quantitative F_{cost} comparisons are presented for all strain amplitude and frequency combinations in Table 6. Table 6 shows that the smallest average value of F_{cost} is achieved with the enhanced thixotropic effects. In Fig. 10e–h the shape of the viscous projections is more linear at the higher strain amplitudes, indicating an almost purely viscous signal. Since the evolving rouleaux structures, and various proteins are held together very weakly and only at lower shear rates, we see that at high strain amplitudes the blood behaves almost like a pure liquid. To demonstrate the ability of the enhanced thixo-EVP models above in Figs. 10,11 and below in Fig. 12 we demonstrate the ability of the model to predict both LAOS and UD-LAOS at least qualitatively, with varying degrees of accuracy. All the hallmark features are shown of qualitative agreement. It is seen that at higher frequencies and strain amplitudes the enhanced *thixo*-EVP models struggle more and more for qualitative agreement. The elastic projections show the stress data and prediction on the y-axis, while the strain is shown on the x-axis. This Lissajous-Bowditch projection shows how the elasticity varies with stress; while the viscous projections show how the viscosity varies with the stress. One can see at the higher values of the strain amplitudes the blood is behaving more and more like a liquid as the viscous projections are more and more closely resembling closed diagonal lines. At the lower strain amplitudes there is more noise in the data because human blood behaves very much like water, with an analogously small viscosity. The EVP only model predictions are not able to qualitatively capture the behavior of the elastic or viscous projections at all combinations of frequency or strain amplitude as well as the models with the enhanced thixotropy modifications. This shows that at lower shear rates there is a microstructure that forms in human blood, that is fleeting, and not seen at higher shear rates. Additionally, it has been shown in the literature that the oscillating motion of the rheometer contributes to breakdown of

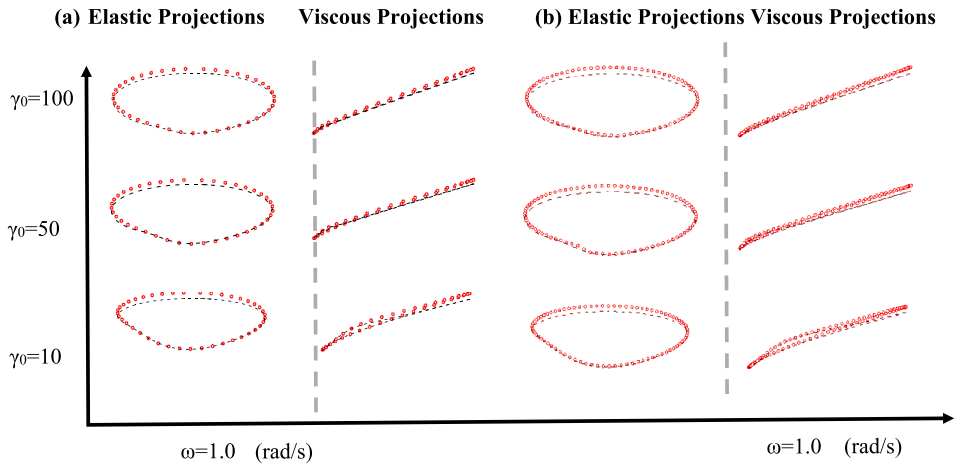


Fig. 12. (a) Elastic and viscous uni-directional LAOS projection (points) and predictions with enhanced thixo-EVP (vers.1) model; (b) Elastic and viscous uni-directional LAOS projection (points) and predictions with enhanced thixo-EVP (vers.2) model of $\gamma_0 = 1(-); \omega = 1(rad/s)$ with Donor 3. (*The individual projections x-axis are the oscillatory contributions to strain and shear rate respectively) [42].

Table 7
Overall comparison over all experimental fits/predictions, over Donors 1–4 (F_{cost} of steady state is non-dimensional).

F_{cost} (Pa)	EVP-only (Stickel et al.)	IKH	ethixo-EVP (vers. 1)	ethixo-EVP (vers. 2)	Varchanis et al.
SS*	0.0310	0.0310	0.0107	0.0107	0.0236
Step Up/ Down	0.000859	0.000696	0.000598	0.000533	0.00112
Ramps	0.00049	0.00043	0.00048	0.00042	0.00093
SAOS	0.00210	0.00217	0.00264	0.00158	0.00348
LAOS	0.00183	0.00161	0.00099	0.00122	0.00408
UDLAOS	0.00095	0.00195	0.00172	0.00168	0.00575
Cessation*	0.00055	0.00103	0.00059	0.00076	0.00182
	0.0068	**0.0079**	**0.0070**	**0.0062**	**0.0172**

structure [21].

Not only has the thixotropic model enhancements allowed qualitative and quantitative evidence of human blood's microstructure but has motivated a continued analysis of this structure. This will simultaneously allow for the prediction of microstructure using the enhanced thixotropic EVP models under LAOS flow and correlating this with the evolution of the elastic and viscous nature of the blood.

Fig. 12 shows similar results as were shown in Fig. 10, that as we increase the strain amplitude there is a more fully developed viscous signal. This is expected since at high strain amplitudes the microstructure and solid like contribution is more broken down. The new enhanced thixo-EVP models can qualitatively and semi-quantitatively predict both LAOS and UD-LAOS, therefore justifying the increase in number of model parameters required to accurately model the effects of human blood's microstructure consisting of rouleaux [8,9]. Table 7 shows the average value, over four sets of Donors, of the F_{cost} for the steady state and all transient flows, which communicates that for human blood thixo-elasto-visco-plastic models are most appropriate. Average AIC and BIC comparisons are shown in the Supplemental Material.

6. Conclusions

We have shown that with recasting of the EVP only model by Stickel et al. [14]in one dimensional form, with a thixotropic parameter to account for the build-up and breakdown of rouleaux (a.k.a. microstructure) of human blood during transient flow is more accurate. The EVP only model was based on a three-dimensional, tensor construct by Saramito [15,40,41]. The new insight is as follows: we are now able to better predict and interpret the complex response of human blood to changes in shear rate for the simple step up and step down in shear rate experiments, as well as the more complicated triangle ramp, cessation flow and LAOS and UD-LAOS with a lower parameter, simple thixotropic construct. This truly shows that by incorporating two separate sets of time scales for human blood both viscoelastic and thixotropic (including the time scales of structure breakage from shear, and structure build up from shear and Brownian motion) we can better capture the actual rheological behavior of the blood. When modeling a thixotropic material, one must consider using the best thixotropy modeling physics, thusly justifying the increase in number of model parameters. However, this allows more accurate predictions of the thixotropic time scales and thixotropic response. This is helpful because it shows that rheologists must adequately factor in all known material time scales for effective and accurate models. In addition to this, the time scale of the evolution of the thixo-elasto-visco-plastic properties of human blood give more insight into the evolution of the rouleaux. To accurately model human blood flow using CFD software one must accurately model the rheological properties as the blood flow develops in arteries. This requires accurate rheological models that directly tie evolving microstructure and viscosity to the three-dimensional flow field as shown by Apostolidis et al. [39–41]. In addition, we envision the use of rheological models to aid in the correlation of rheological parameters to physiological parameters in statistically meaningful ways, whereby the physiological data from the lab shown in Table 1 for Donor 3 could be used in a meaningful way to determine overall blood health in the future, but only with the accurate thixo-elasto-visco-plastic models to fit rheological parameters. With the aid of the thixotropic enhancements in conjunction with parametric analysis one can now better understand the complexities of the mechanical properties of human blood in terms of relationships between evolving structure, elastic, and viscous properties. Interestingly due to the 'lightly structured' aspects of human blood at low shear rates, the 'back-stress' approach of Dimitriou et al., Wei et al. and Varchanis et al. did not adequately model the behavior of the rouleaux. The standard *3-term* thixotropy modeling allowed the ethixo-EVP models to capture the transient rheological experiments much better. The Varchanis et al. inspired ethixo-EVP model was able to predict the transient rheological blood flows as well as shown in Table 7. Table S4 of the Supplemental Material shows additional results comparing Donors 1-4 [42–45].

Moving forward we aim to apply the thixotropic modeling to more physiological varied samples of human blood to look at how various physiological measures such as cholesterol, hematocrit, fibrinogen, etc. influence the microstructure, elastic, and viscous properties of human

blood. We will endeavor to tie together not only the experimental protocol to the microstructure and mechanical properties of blood but also look for correlations between these properties and physiological conditions.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jnnfm.2021.104503.

Appendix A. ethixo-EVP tensor derivation

Following the technique to transform a scalar model into tensor form outlined in Chapter 7, of the Dynamics of Polymeric Materials text by Bird, Armstrong and Hassager [38], starting with Eq. (11), then making the following substitution for $\dot{\gamma}$, and σ respectively

$$\dot{\gamma} = \underline{\underline{\gamma}}_{(1)} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \dot{\gamma}_{yx}, \tag{A1}$$

$$\underline{\underline{\tau}} = \begin{pmatrix} \tau_{xx} & \tau_{yx} & 0 \\ \tau_{yx} & \tau_{yy} & 0 \\ 0 & 0 & \tau_{zz} \end{pmatrix}. \tag{A2}$$

With the derivative of the stress given by the upper convected derivative of the stress tensor as follows from Table C.1 of [38]

$$\dot{\sigma} = \underline{\underline{\tau}}_{(1)} = \frac{\partial}{\partial t}\begin{pmatrix} \tau_{xx} & \tau_{yx} & 0 \\ \tau_{yx} & \tau_{yy} & 0 \\ 0 & 0 & \tau_{zz} \end{pmatrix} - \begin{pmatrix} 2\tau_{yx} & \tau_{yy} & 0 \\ \tau_{yy} & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \dot{\gamma}_{yx} \tag{A3}$$

By substituting Eqs. (1)–(3) into Eq. (11), we obtain the ethixo EVP (vers.1) for each of the components:

$$yx: \frac{1}{G}\left(\frac{\partial \tau_{yx}}{\partial t} - \tau_{yy}\dot{\gamma}_{yx}\right) + max\left(0, \frac{|\tau_{yx}| - \lambda\sigma_{y0}}{(\lambda\eta_{st} + \eta_{\infty})|\tau_{yx}|}\right)\tau_{yx} = \dot{\gamma}_{yx} \tag{A4}$$

$$xx: \frac{1}{G}\left(\frac{\partial \tau_{xx}}{\partial t} - 2\tau_{yx}\dot{\gamma}_{yx}\right) + max\left(0, \frac{|\tau_{xx}| - \lambda\sigma_{y0}}{(\lambda\eta_{st} + \eta_{\infty})|\tau_{xx}|}\right)\tau_{xx} = 0 \tag{A5}$$

$$yy: \frac{1}{G}\frac{\partial \tau_{yy}}{\partial t} + max\left(0, \frac{|\tau_{yy}| - \lambda\sigma_{y0}}{(\lambda\eta_{st} + \eta_{\infty})|\tau_{yy}|}\right)\tau_{yy} = 0 \tag{A6}$$

$$zz: \frac{1}{G}\frac{\partial \tau_{zz}}{\partial t} + max\left(0, \frac{|\tau_{zz}| - \lambda\sigma_{y0}}{(\lambda\eta_{st} + \eta_{\infty})|\tau_{zz}|}\right)\tau_{zz} = 0 \tag{A7}$$

Other convected derivatives can be incorporated instead of the upper convected, the evolution of the structure parameter is not changed. We note here that the steady state solution is ill-defined if one tries to evaluate directly by equating the time derivatives to zero. Perhaps the best way to evaluate the steady state is by solving the full time-dependent equations starting from rest (when all the stress components are zero) and considering the limiting value for any given shear rate at time going to infinity. We also acknowledge that one may choose to substitute other criteria for yielding in the numerator of the conditional, for example the Von Mises yield criteria: $|\sqrt{3}\tau_{12}| - \lambda\sigma_{y0}$ [27]. Some results of the tensor version of the upper convected ethixo EVP model are shown in the Supplementary Material, Table S1. An analogous procedure can be followed for version 2 of the ethixo EVP model.

References

[1] G. Lowe, Blood rheology in arterial disease, Clin. Sci. 71 (1986) 137–146.
[2] G. Cokelet, Hemorheology and hemodynamics, Colloqu. Seq. Integrat. Syst. Physiol. Molecule Funct. (2011) 1–17.
[3] D.A. Reasor, J.R. Clausen, C.K. Aidun, Rheological characterization of cellular blood in shear, J. Fluid Mech. 726 (2013) 497–498, https://doi.org/10.1017/jfm.2013.229.
[4] E. Merrill, Rheology of blood, Physiol. Rev. 49 (4) (1969).
[5] A. Apostolidis, A. Beris, Modeling the blood rheology in steady-shear flows, J. Rheol. 58 (3) (2014) 607–633.
[6] A. Apostolidis, A. Beris, The effect of cholesterol and triglycerides on the steady state rheology of blood, Rheo. Acta (1) (2016) 1–13.
[7] A. Apostolidis, M. Armstrong, A. Beris, Modeling of human blood rheology in transient shear flows, J. Rheol. 59 (2015) 275–298.

[8] J. Horner, M. Armstrong, N. Wagner, A. Beris, Investigation of blood rheology under steady and unidirectional large amplitude oscillatory shear, J. Rheol. 62 (2018) 577–591, a.
[9] J. Horner, M. Armstrong, N. Wagner, A. Beris, Measurements of human blood viscoelasticity and thixotropy under steady and transient shear and constitutive modeling thereof, J. Rheol. 63 (2019) 799–813.
[10] L. Moreno, F. Calderas, G. Sanchez-Olivares, L. Medina-Torres, A. Sanchez-Solis, O. Manero, Effect of cholesterol and triglycerides levels on the rheological behavior of human blood, Korea-Aust. Rheol. J. 27 (1) (2015) 1–10.
[11] S. Varchanis, Y. Dimakopoulos, C. Wagner, J. Tsamopoulos, How viscoelastic is human blood plasma? Soft Matter 14 (2018) 4238–4251.
[12] J. Mewis, N. Wagner, Colloidal Suspension Rheology, Cambridge University Press, 2012, pp. 25–30.
[13] K. Dullaert, J. Mewis, A structural kinetics model for thixotropy, J. Non Newton. Fluid Mech. 139 (2006) 21–30.
[14] J. Stickel, J. Knutsen, M. Liberatore, Response of elastoviscoplastic materials to large amplitude oscillatory shear flow in the parallel-plate and cylindrical-Couette geometries, J. Rheol. 57 (6) (2013) 1569–1595.
[15] P. Saramito, A new elastoviscoplastic model based on the Herschel-Bulkley viscoplastic model, J. Non Newton. Fluid Mech. 158 (2009) 154–161.
[16] P. de Souza Mendes, Modeling the thixotropic behavior of structure fluids, J. Non Newton. Fluid Mech. 164 (2009) 66–75.
[17] P. de Souza Mendes, Thixotropic elasto-viscoplastic model for structured fluids, Soft Matter 7 (2011) 2471–2483.
[18] P. de Souza Mendes, R. Thompson, A unified approach to model elasto-viscoplastic thixotropic yield-stress and apparent yield stress fluids, Rheo. Acta 52 (673) (2013) 694.
[19] A. Mujumdar, A. Beris, A. Metzner, Transient phenomena in thixotropic systems, J. Non Newton. Fluid Mech. 102 (2002) 157–178.
[20] B. Blackwell, R. Ewoldt, A simple thixotropic-viscoelastic constitutive model produces unique signatures in large amplitude oscillatory shear (LAOS), J. Non Newton. Fluid Mech. (2014) 208–209, 27-41.
[21] M. Armstrong, A. Beris, S. Rogers, N. Wagner, Dynamic shear rheology of a thixotropic suspension: comparison of an improved structure-based model with large amplitude oscillatory shear experiments, J. Rheol. 60 (2016) 433–450.
[22] M. Armstrong, A. Beris, N. Wagner, Dynamic shear rheology and structure kinetics modeling of a thixotropic carbon black suspension, Rheo. Acta 56 (10) (2017) 811–824.
[23] Y. Wei, M. Solomon, R. Larson, Quantitative nonlinear thixotropic model with stretched exponential response in transient shear flows, J. Rheol. 62 (2018) 321–342.
[24] R. Larson, Y. Wei, A review of thixotropy and its rheological modeling, J. Rheol. 63 (2019) 477–501.
[25] C. Dimitriou, R. Ewoldt, G. McKinley, Describing and prescribing the constitutive response of yield stress fluids using large amplitude oscillatory shear stress (LAOStress), J. Rheol. 57 (1) (2013) 27–70.
[26] C. Dimitriou, G. McKinley, A comprehensive constitutive law for waxy crude oil: a thixotropic yield stress fluid, Soft Matter 10 (35) (2014) 6619–6644.
[27] S. Varchanis, G. Makrigiorgos, P. Moschopoulos, Y. Dimakopoulos, J. Tsamopoulos, Modeling the rheology of thixotropic elasto-viscoplastic materials, J. Rheol. 63 (4) (2019) 609–639.
[28] P. Sousa, K. Carneiro, R. Vaz, A. Carejo, F. Pinho, M. Alves, M. Oliveira, Shear viscosity and nonlinear behavior of whole blood under large amplitude oscillatory shear, Biorheology 50 (5-6) (2013) 269–282.
[29] M. Clarion, M. Deegan, T. Helton, J. Hudgins, N. Monteferrante, E. Ousley, M. Armstrong, Contemporary modeling of human blood rheology, Rheol. Acta 57 (2) (2018) 141–168.
[30] M. Armstrong, J. Horner, M. Clark, M. Deegan, T. Hill, C. Keith, L. Mooradian, Evaluating rheological models for human blood using steady state, transient and oscillatory shear predictions, Rheol. Acta 57 (2018) 705–728.
[31] M. Bureau, J. Healy, D. Bourgoin, M. Joly, Rheological hysteresis of blood at low shear rate, Biorheology 17 (1980) 191–203.
[32] J. Giacomin, J. Dealy, Large amplitude oscillatory shear, in: AA Collyer (Ed.), Techniques in Rheological Measurement, Springer Dordrecht, 1993.
[33] K. Baskurt, M. Boynard, G. Cokelet, P. Connes, B. Cooke, S. Forconi, F. Liao, M. Hardeman, F. Jung, H. Meiselman, G. Nash, N. Nemeth, B. Neu, B. Sandhagen, S. Shin, G. Thurston, J. Wautier, New guidelines for hemorheological laboratory techniques, Clin. Hemorheol. Microcirc. 42 (2009) 75–97.
[34] J. Horner, A. Beris, D. Woulfe, N. Wagner, Effects of ex vivo aging and storage temperature on blood viscosity, Clin. Hemorheol. Microcirc. 70 (2018) 155–172.
[35] M. Armstrong, N. Wagner, A. Beris, An adaptive parallel tempering method for the dynamic data-driven parameter estimation of nonlinear models, AIChE J. 63 (2017) 1937–1958.
[36] H. Akaike, A new look at the statistical model identification, IEEE Trans. Autom. Control AC 19 (6) (1974) 716–723.
[37] J. Freund, R. Ewoldt, Quantitative rheological model selection: Good fits versus credible models using Bayesian inference, J. Rheo. 59 (3) (2015) 667–701.
[38] R. Bird, R. Armstrong, O. Hassager, Dynamics of Polymeric Liquids: Fluid Mechanics, 2nd Ed., 1, United States. John Wiley and Sons Inc, 1987.
[39] A. Apostolidis, A. Moyer, A. Beris, Non-Newtonian effects in simulations of coronary arterial blood flow, J. Non Newton. Fluid Mech. 233 (2016) 155–165.
[40] P. Saramito, A new constitutive equation for elastoviscoplastic fluid flows, J. Non Newton. Fluid Mech. 145 (1) (2007) 1–14.
[41] N. Thien, R. Tanner, A new constitutive equation derived from network theory, J. Non Newton. Fluid Mech. 2 (4) (1977) 353–365.
[42] dataset M. Armstrong, J. Horner, Rheology data of human blood JUN18; SAOS, Step up/down, LAOS, UDLAOS, Cessation of flow with ARESG2 Mendeley Data, v1, Mendeley Data, 2020, https://doi.org/10.17632/948ffnypjs.1.
[43] dataset M. Armstrong, J. Horner, Rheology data of human blood JUL18; SAOS, Step up/down, LAOS, UDLAOS with ARESG2 Mendeley Data, v1, 2020, https://doi.org/10.17632/s8w6s6f68b.
[44] dataset M. Armstrong, DEC17 ARESG2 Human Blood Rheology: SS, SAOS, LAOS, STEPS, etc., Mendeley, 2020 https://doi.org/10.17632/d43r5ydpx8.1.
[45] dataset M. Armstrong, J. Horner, FEB18 Blood Rheology ARESG2 SS SAOS LAOS STEP, Mendeley, 2020, https://doi.org/10.17632/jdzpx6hy6r.1.